Türkiye at the Crossroads:
Climate Change Impacts and Internal Migration Determinants

Sedat Ors*

July 2026

1 Introduction

Climate change is increasingly recognized as an important driver of human mobility, reshaping how researchers and policymakers understand population movements in response to environmental stress. A growing body of evidence indicates that climate change is associated with rising temperatures, changing precipitation patterns, and sea-level rise (IPCC, 2022; Nordhaus, 2013; Nordhaus, 2019). These environmental changes are expected to increase the frequency, intensity, and severity of climate-related hazards, including droughts, floods, wildfires, and storms (Halliday, 2006; Koubi, Stoll, and Spilker, 2016; Beine and C. R. Parsons, 2017; Murray-Tortarolo and Martínez Salgado, 2021; IPCC, 2022). As climatic shocks alter agricultural productivity, livelihoods, and living conditions, affected populations may respond by migrating temporarily or permanently (Carleton, Jina, et al., 2022). The relationship between climate change and migration, however, remains complex, reflecting the diverse economic, social, and environmental mechanisms through which climate variability influences human mobility (Tol, 2002; Piguet, Pecoud, and De Guchteneire, 2011; Beine and Jeusette, 2021; M. Moore and Wesselbaum, 2023; Thorn et al., 2023). Understanding these dynamics is therefore critical for evaluating the socioeconomic consequences of climate change and designing effective adaptation strategies. As Cifci and Oliver (2018) note, addressing climate change involves balancing urgent adaptation needs with substantial economic costs and potential impacts on development.

*PhD, School of Economics, Georgia Institute of Technology, Atlanta, GA, USA. Email: sors3@gatech.edu, ORCID: 0009-0006-1953-1604.

Early work by El-Hinnawi (1985) introduced the concept of environmental migrants, referring to individuals who move in response to sudden or gradual environmental change.[1] Subsequent research emphasizes that migration responses to environmental shocks depend on complex interactions among environmental conditions, economic opportunities, demographic pressures, and political institutions (Black et al., 2011; Piguet, Pecoud, and De Guchteneire, 2011; Coniglio and Pesce, 2015; Cattaneo, Beine, et al., 2019; Thorn et al., 2023). These developments have also generated increasing discussion of the concept of "climate refugees," highlighting how environmental shocks may lead to large-scale displacement (Black, 2001; Brown, 2008; Nawrotzki et al., 2015; Helbling and Meierrieks, 2021; IPCC, 2022). Estimates of future climate-induced migration vary considerably. One influential estimate by Myers (2002) suggested that up to 200 million people could be displaced by 2050, while more recent projections indicate that more than 143 million people in Sub-Saharan Africa, South Asia, and Latin America may migrate internally because of climate change (Rigaud et al., 2018).

Climate change may also influence migration indirectly through its effects on economic conditions and conflict. Climatic shocks can reduce agricultural productivity while increasing poverty and economic vulnerability (Schlenker and Roberts, 2009; Cai et al., 2016; Cattaneo and Peri, 2016). Extreme temperatures and prolonged droughts can further intensify these pressures, contributing to social instability and displacement (Kelley et al., 2017; Selby et al., 2017; Chen and Mueller, 2019). A prominent example is the severe drought that affected Syria between 2007 and 2010. Kelley et al. (2017) argue that this prolonged drought contributed to agricultural collapse, rural displacement, and economic hardship, factors that intensified the Syrian civil conflict and generated large refugee flows.

Türkiye has been directly affected by these regional dynamics. The country hosts one of the world's largest refugee populations, including approximately 3.5 million Syrians according to the Presidency of Migration Management (PMM).[2] The arrival of such a large refugee population has

[1]Environmental migrants are individuals or groups who, because of environmental changes that adversely affect their living conditions, are forced to leave their habitual residence either temporarily or permanently. The terminology remains debated, particularly because the term "refugee" has a specific legal meaning under international law.

[2]Official statistics available at: `https://en.goc.gov.tr/`

placed additional pressure on housing markets, public services, and local labor markets in several regions of the country. These demographic pressures may, in turn, influence the mobility decisions of the local population, generating additional internal migration flows (Erdoğan and Cantürk, 2022). Migration may also reshape social and cultural interactions between host communities and migrant populations through compositional change and cultural diffusion (Rapoport, Sardoschau, and Silve, 2022). Together, these mechanisms suggest that climate-related shocks may generate indirect migration effects, whereby displacement in one region creates additional migration pressures elsewhere (Joseph and Wodon, 2013).

Against this background, I examine how climatic conditions and refugee-related demographic pressures influence internal migration patterns in Türkiye between 2008 and 2024.[3] I focus on internal migration because it represents the primary adjustment mechanism through which households respond to regional climatic and economic conditions. International migration, by contrast, is subject to substantial institutional and policy barriers that constrain cross-border mobility (McCallum, 1995; Anderson and Wincoop, 2003; Chen and Mueller, 2019). Using province-level migration flows between origin and destination provinces, I examine how climatic, economic, demographic, and geographic factors jointly shape migration decisions.

To capture heterogeneity in climatic exposure, I conduct the empirical analysis using three complementary samples. First, I examine the full national migration system covering all 81 provinces of Türkiye. Second, I analyze a subsample of 30 provinces with relatively greater exposure to climatic variability and environmental stress.[4] Third, I focus on a narrower subsample of 17 drought-affected provinces identified using the Bagnouls–Gaussen aridity index developed by Cebeci et al. (2019).[5] This multi-sample design allows me to evaluate whether the relationship between climatic conditions and migration is stronger in regions that are more vulnerable to climatic stress. A complete list of provinces included in the climate-exposed and drought-exposed

[3]The analysis begins in 2008 because consistent migration data are available from the Turkish Statistical Institute.

[4]The selection of these provinces is based on climatic exposure, their geographic location in southern Türkiye, and urban–rural characteristics.

[5]The Bagnouls–Gaussen aridity index measures climatic dryness by comparing temperature and precipitation and is widely used to identify drought-prone regions.

subsamples is provided in Appendix, Table A.1.

My empirical framework combines the Random Utility Model (RUM) with a gravity-type migration model (Backhaus, Martinez-Zarzoso, and Muris, 2015; Beine and C. R. Parsons, 2017). The gravity framework captures how climatic, economic, demographic, and geographic factors at both the origin and destination shape migration flows, while the RUM provides the microeconomic foundation for destination choice based on differences in expected utility. I estimate the model using three complementary approaches: a three-way fixed-effects specification, the Poisson Pseudo-Maximum Likelihood (PPML) estimator, and the Hausman–Taylor estimator. Together, these approaches account for unobserved heterogeneity, zero migration flows, and potential endogeneity arising from time-invariant regressors (Hausman and Taylor, 1981; Santos Silva and Tenreyro, 2006; Wooldridge, 2010).

This study provides one of the first province-level panel analyses of climate and internal migration in Türkiye. The results consistently highlight the roles of climatic, economic, and demographic factors in shaping internal migration. Higher annual average temperatures in origin provinces are associated with greater out-migration, while the temperature-bin specification indicates that migration rates increase sharply once summer maximum temperatures exceed approximately 32 Celcisus(°C). Precipitation also appears to influence migration, particularly in drought-prone regions, while climatic conditions in destination provinces affect migration through an amenity channel. Economic conditions remain important, as higher income at origin relaxes mobility constraints and higher income at destination attracts migrants. Finally, refugee inflows are associated with higher internal migration flows in several specifications, suggesting that demographic pressures may amplify population movements through secondary displacement.

The remainder of this chapter is organized as follows: section 2 reviews the relevant literature, and section 3 provides background information on Türkiye, section 4 presents the methodological framework, while section 5 describes the data and summary statistics. section 6 outlines the identification strategy, section 7 reports the empirical results, section 8 discusses the findings, and section 9 concludes.

2 Literature Review

Nordhaus (2019) argues that climate change has become one of the defining economic and environmental challenges of recent decades. Although natural climate variability has generated temperature fluctuations throughout history, recent warming trends are largely attributed to human activity. Rising temperatures alter precipitation patterns, increase the frequency of extreme weather events, reduce water availability, and affect other environmental conditions that influence economic activity and human welfare. Climate projections further indicate substantial increases in global temperatures over the coming century in the absence of significant mitigation efforts (Nordhaus and Boyer, 2003; Nordhaus, 2013). These long-run climatic trends provide important context for understanding how environmental change may influence migration decisions. Appendix Figure A.1 presents a long-run reconstruction of global temperatures based on Antarctic ice-core data.

A growing body of empirical research examines the relationship between climate change and migration, showing how climatic conditions shape human mobility across a wide range of contexts (Beine and C. Parsons, 2015; Backhaus, Martinez-Zarzoso, and Muris, 2015; Berlemann and Steinhardt, 2017; Berlemann and Tran, 2020; Beine and Jeusette, 2021). Much of this work finds that rising temperatures and shifting precipitation patterns are central determinants of migration decisions (Backhaus, Martinez-Zarzoso, and Muris, 2015; Cai et al., 2016; Cattaneo and Peri, 2016). These same changes also increase the frequency and severity of natural hazards—including droughts, floods, sea-level rise, and wildfires—which in turn reshape migration flows (Suhrke and Hazarika, 1993; Halliday, 2006; Kharin et al., 2007; Millock, 2015; IPCC, 2022).

Migration responses to climatic change, however, are never purely environmental; they are mediated by economic and social conditions. Classical and contemporary migration theory alike emphasize that mobility reflects a combination of environmental pressures, economic opportunities, and institutional constraints (Ravenstein, 1885; Black, 2001; Thorn et al., 2023). As Piguet, Pecoud, and De Guchteneire (2011) argue, climate change affects both the willingness and the ability to migrate, which makes its direct and indirect effects difficult to disentangle. At one extreme, rising sea levels have rendered low-lying areas such as Tuvalu increasingly uninhabitable, so that migration becomes a direct response to environmental change (Campbell and Warrick, 2014). More commonly, climate operates indirectly, through declines in agricultural productivity and household income that reshape both the incentives and the means to move (Coniglio and Pesce, 2015; Beine and C. R. Parsons, 2017).

Among specific drivers, temperature has received the most attention. Higher temperatures worsen living conditions and, in many settings, raise out-migration (Bohra-Mishra, Oppenheimer, and Hsiang, 2014; Beine and C. R. Parsons, 2017), with the response often nonlinear and concentrated at extreme heat (Missirian and Schlenker, 2017; Carleton, Hsiang, and Burke, 2016). These effects are strongly heterogeneous across income levels: Cattaneo and Peri (2016) find that higher temperatures raise emigration from middle-income countries but not from the poorest, where liquidity constraints can trap potential migrants—a mechanism reinforced by evidence on resource-constrained immobility (Benveniste, Oppenheimer, and Fleurbaey, 2022). The sign is not fully settled, however, as Beine and C. R. Parsons (2017) report higher temperatures deterring emigration from middle-income origins, underscoring how sensitive these estimates are to specification.

Precipitation plays a complex role, largely operating through agriculture. Rainfall shocks shape rural livelihoods and migration in sub-Saharan Africa (Barrios, Bertinelli, and Strobl, 2006) and have been used to explain out-migration from rural Mexico (Munshi, 2003), whereas in rural Ecuador, higher rainfall has been associated with reduced internal and international movement (C. L. Gray, 2009). Disaggregated evidence from Vietnam is particularly instructive. Berlemann and Tran (2020) find that droughts and worsening flood trends increased internal migration, with movements initially being temporary but becoming increasingly permanent as climatic conditions deteriorated. By contrast, typhoons had no significant effect.

Meta-analyses and comparative studies further highlight substantial heterogeneity in climate-migration relationships. Beine and Jeusette (2021) demonstrate that estimated climate effects vary considerably across studies due to differences in methodology, climate measures, migration definitions, and geographic context. A broader empirical literature similarly documents substantial variation in migration responses across countries and regions (Beine and C. Parsons, 2015; Millock, 2015; Ruiz, 2017; Berlemann and Steinhardt, 2017; M. Moore and Wesselbaum, 2023). More generally, countries in the Global South tend to face greater exposure to climate risks and lower adaptive capacity than countries in the Global North, contributing to migration toward wealthier and less climate-vulnerable destinations (Sasser, 2010; Piguet, Kaenzig, and Guelat, 2018; Cattaneo, Beine, et al., 2019). The projected scale of future climate-related migration is substantial. Myers (2002) estimated that more than 200 million people could be displaced by climate change by 2050, while Rigaud et al. (2018) projected that over 143 million individuals may migrate internally within developing regions by mid-century.

Climate change may also affect migration through broader economic and social mechanisms. Climatic shocks can reduce agricultural production, increase poverty, and heighten economic vulnerability (Cai et al., 2016; Cattaneo and Peri, 2016; Beine and C. R. Parsons, 2017). These pressures may be intensified by extreme temperatures and recurrent droughts (Roberts, Schlenker, and Eyer, 2013), which can contribute to conflict and displacement (Koubi, Stoll, and Spilker, 2016; Kelley et al., 2017; Selby et al., 2017; Chen and Mueller, 2019). A prominent example is the Syrian drought of 2007–2010. Kelley et al. (2017) argue that climate conditions contributed to agricultural collapse and economic instability, factors that helped intensify conflict and generate large refugee flows.

Climate-related migration may also create secondary effects in receiving regions, including increased pressure on labor markets, public services, and social cohesion (Joseph and Wodon, 2013; Strobl and Valfort, 2015; Thiede, C. Gray, and Mueller, 2016; Dallmann and Millock, 2017; Ruiz, 2017; C. Gray, Hopping, and Mueller, 2020; Zouabi, 2021). In addition, migration can reshape social and cultural interactions through compositional change and cultural diffusion (Rapoport, Sardoschau, and Silve, 2022).

Despite this extensive global literature, evidence for Türkiye remains limited, and the climate and migration literatures have largely developed in isolation. On the climate side, Cebeci et al.

(2019) measures drought vulnerability across Turkish provinces, while government reports document the impacts of droughts, floods, and wildfires. Related studies examine environmental change, urban development, and climate-related vulnerabilities within the Turkish context (Ozceylan and Coskun, 2012; Benli et al., 2018). On the migration side, Erdoğan and Cantürk (2022) focuses on international migration flows into Türkiye. What remains largely unexplored is how climatic conditions influence *internal* migration within the country. This study addresses that gap by bringing climate variability and internal migration into a unified empirical framework for Türkiye.

This paper makes several contributions. First, it offers one of the first comprehensive analyses of internal migration in Türkiye to consider climatic, economic, and demographic factors jointly within a single empirical framework. Whereas existing work typically examines climate impacts or migration in isolation, this study integrates the two and explicitly models conditions at both origin and destination locations.

Second, it contributes to the literature on climate-induced migration by examining nonlinear temperature effects. The results indicate that out-migration increases sharply once temperatures exceed a critical threshold. This pattern is consistent with recent evidence on the nonlinear impacts of extreme heat on human outcomes (Carleton, Hsiang, and Burke, 2016; Missirian and Schlenker, 2017) and underscores the importance of accounting for temperature extremes rather than relying solely on average climate measures.

Third, the analysis employs multiple estimation approaches, including three-way fixed effects, PPML, and Hausman-Taylor estimation. Comparing results across these estimators helps address concerns related to zero migration flows, time-invariant frictions such as distance, and unobserved heterogeneity.

Finally, the study incorporates refugee inflows as an additional channel influencing internal migration. Given Türkiye's role as a major host country for refugees, accounting for refugee presence provides insight into how demographic pressures may interact with climatic and economic conditions to shape population movements within the country.

3 Overview of Türkiye

Türkiye, situated between 36° and 42° north latitude and 26° and 45° east longitude, lies at the crossroads of Europe and Asia, exposing it to diverse climatic, geographic, and socio-economic conditions that make it particularly sensitive to the impacts of climate change. With a population of approximately 86 million and a relatively young median age of about 32 years, the country is highly urbanized, as nearly 75% of people reside in metropolitan areas such as Istanbul, Ankara, and Izmir, largely driven by rural to urban migration in search of employment, education, and improved living standards. Rural areas, however, remain dependent on agriculture and livestock, increasing their vulnerability to drought and extreme weather events. In addition, Türkiye hosts nearly 5 million refugees, of whom approximately 3.5 million are from Syria, reflecting its central role in regional migration dynamics.[6]

Türkiye's geography features sharp contrasts in topography and climate over relatively short distances, producing substantial regional disparities in socio-economic conditions. The country experiences three dominant climate types: Mediterranean in the southwest, Black Sea in the north, and Continental across the interior and eastern regions. Much of the Black Sea, Mediterranean, and Eastern Anatolia regions is mountainous, with east–west ranges reaching considerable elevations. This geographic configuration generates strong variations in climate, livelihoods, and lifestyles across the country (Tekin, 2023).

Türkiye is part of the Mediterranean Basin, one of the regions most vulnerable to climate change. Since the 1980s, the country has experienced notable shifts in temperature and precipitation, with a clear trend toward a warmer and drier climate (Türkeş and Tatlı, 2009; Türkeş, 2012; Demircan et al., 2017). Climate projections indicate that these patterns will intensify in the coming decades (Bozkurt and Sen, 2013; Bayram and Öztürk, 2021). Türkiye's precipitation climatology is strongly shaped by the North Atlantic Oscillation (NAO): positive phases of the NAO are linked to severe droughts, while negative phases bring increased rainfall (Türkeş and Erlat, 2005). With climate change, the positive intensity of this oscillation—and thus the risk of drought

[6] Official statistics available at: https://en.goc.gov.tr/

in Türkiye—is expected to rise (Visbeck et al., 2001).

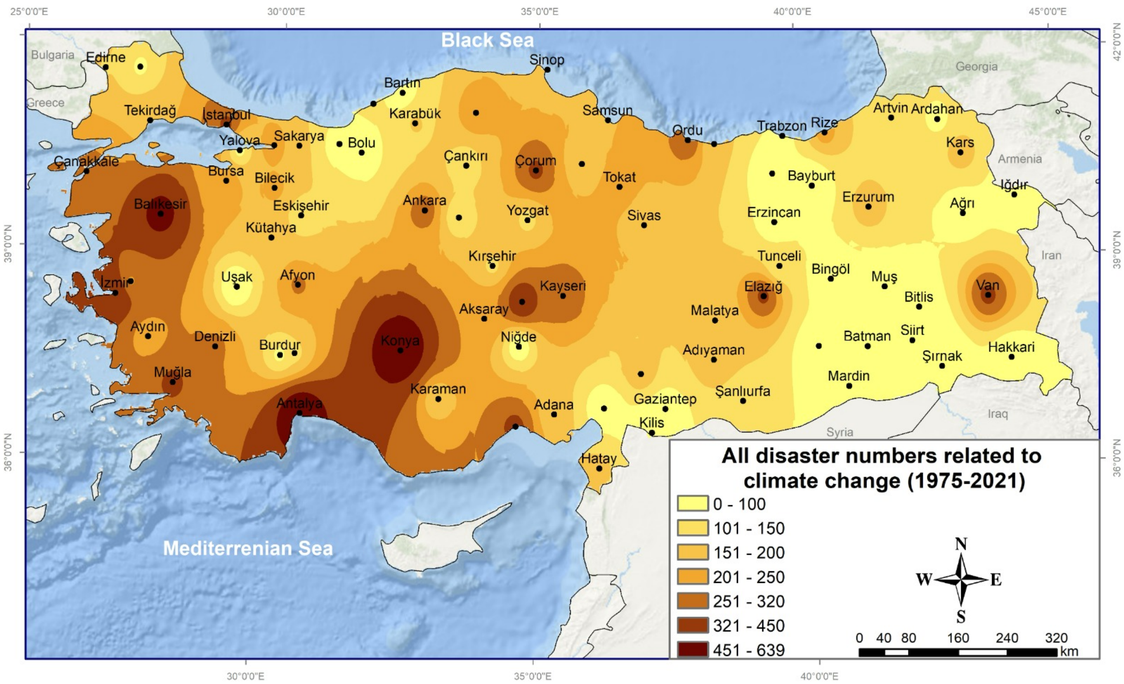

Figure 1: Total Climate Events in Türkiye from 1975 to 2021

Note: The figure shows the total number of climate-related disaster events, including droughts, extreme temperatures, heatwaves, floods, wildfires, and storms. Data Source: Tekin (2023)

Across Türkiye, climate-related events have become more frequent in recent decades. Figure 1 shows the spatial distribution of recorded climate-related events between 1975 and 2021.[7] Recorded events are concentrated in the southern regions, reflecting their greater exposure to climatic hazards. Part of this concentration may also reflect higher population density and infrastructure, since events are more likely to be recorded where they affect people and assets. The map spans droughts, extreme temperatures, heatwaves, floods, wildfires, and storms, with darker shades indicating higher frequency.

Drought, in particular, is poised to become one of the most consequential climatic hazards for Türkiye's future (Tekin, 2023). Apart from the wetter Black Sea region, drought is expected to intensify across Central Anatolia, Southeastern Anatolia, and the Mediterranean, and current assessments indicate that a substantial share of the country's land is at risk of drought and desertification

[7]The figure reports event counts only, not their intensity or consequences; it is nonetheless the most illustrative overview currently available.

(Türkeş, 2012; Bayram and Öztürk, 2021; Serkendiz et al., 2023). Because rural livelihoods in these regions rest on agriculture and livestock, intensifying drought directly threatens incomes and may push affected populations to migrate.

Türkiye also sits at the crossroads of major migration routes. Its location between mid-latitude and tropical-origin pressure systems produces a range of sub-climates, and it has long served as a transit country: migrants from the Middle East and South Asia—Afghans, Iranians, Iraqis, Syrians, and Pakistanis–pass through Türkiye en route to Europe. As natural disasters grow more frequent in these politically and economically fragile regions, Türkiye is likely to keep receiving migrants displaced by drought from the Middle East and Western Asia, and potentially climate migrants from farther afield (Tekin, 2023). Many neighboring countries are at high risk of food insecurity and depend heavily on climate-dependent livelihoods (Krishnamurthy, K. Lewis, and Choularton, 2014), so continued climate stress in the region is likely to sustain these flows. These same pressures, most visibly the Syrian refugee inflows, also bear on internal migration within Türkiye, the focus of this paper.

4 Methodology

In this study, I combine the gravity model with the RUM to analyze the determinants of migration; the RUM supplies the behavioral foundation for individual location choice, and the gravity equation supplies the corresponding empirical specification for bilateral flows. The gravity model in economics originates with Tinbergen (1962) and was developed further in the trade literature by McCallum (1995) and Anderson and Wincoop (2003), who formalized border effects and multilateral resistance, before being applied to labor and migration flows. Beine and C. Parsons (2015) brought the framework to climate-driven international migration, and subsequent work has used it to study migration dynamics while accounting for geographic proximity, economic disparities, and climate variables such as temperature and precipitation (Afifi and Warner, 2008; Backhaus, Martinez-Zarzoso, and Muris, 2015; Beine, Bertoli, and Fernández-Huertas Moraga, 2016). The RUM, in turn, provides a formal framework for an individual's decision to move between provinces, comparing the utility of origin and destination under current and expected future conditions (Coniglio and Pesce, 2015; Beine and C. R. Parsons, 2017; Ruiz, 2017; Freeman and J. Lewis, 2021).

4.1 Random Utility Model

In this section, I establish a mathematical model using RUM (Beine, Docquier, and Özden, 2011; Beine and C. R. Parsons, 2017; Ruiz, 2017; Freeman and J. Lewis, 2021) to determine the conditions under which an individual decides to migrate from one province to another within a country. The RUM evaluates the utility of residing in the origin and destination provinces, accounting for current conditions and expected future conditions.

To define the utility function, I incorporate several factors that influence an individual's decision to migrate, broadly categorized as push and pull factors. As noted by Ravenstein (1885), people migrate to improve their circumstances, driven by social, political, and environmental conditions. Push factors are unfavorable conditions in the origin location, such as high living costs, adverse climate, or limited economic opportunities, while pull factors are attractive features of the destination, such as better employment opportunities, favorable climate, and lower living costs (Rummukainen, 2012).[8] Within this framework, Afifi and Warner (2008) emphasized that hunger, war, and human rights violations serve as extreme push factors driving migration. Similarly, Burzyński et al. (2022) identified real wages, education levels, and migration costs as determinants that can either push individuals away from the origin or pull them toward a destination. Individual risk preferences also shape these dynamics. Becker (1962) argued that "risk-lovers" are more likely to migrate than "risk-averse" individuals, because they are more willing to accept the uncertainty associated with migration. Finally, Rapoport, Sardoschau, and Silve (2022) noted that migration entails not only economic push and pull forces but also cultural ones, as individuals weigh both the loss of familiar environments and the potential cultural opportunities offered by the

[8]Migration decisions are influenced by a wide range of factors, including demographic trends, policy interventions, and personal circumstances. For tractability, however, I restrict my model to the most widely studied and quantifiable determinants: economic, climatic, and social push-and-pull factors.

destination.

An individual decides to migrate from origin province i to destination province j if the expected utility in the destination province exceeds the current utility in the origin province, as shown in eq:Mutilityexceeds. Conversely, when the inequality in eq:Mutilityexceeds does not hold, the individual chooses to stay in the origin location. I model the utility function using expectations for the next period, following the approach of Reuveny and W. H. Moore (2009) and Coniglio and Pesce (2015), who also incorporate expected future utility into their models.[9] However, Cattaneo and Peri (2016) and Beine and C. R. Parsons (2017) employed a two-period model with U_{t+1} for the next period, solving their models over two periods. Additionally, Rappaport (2009) modeled dynamic and steady-state levels over all periods.

$$U_{it} < U_{jt}^{e} \tag{1}$$

The utility in origin city i at time t, U_{it}, is influenced by several factors, including income, private benefits such as family ties or attachment to the origin provinces, daily living costs, and climate-related costs such as extreme temperatures and droughts.

First, I express the utility function in general terms for the origin province in equation eq:Morigincomponents:

$$U_{it}(I_{it}, B_{it}, C_{it}) = I_{it} + B_{it} - C_{it} + \epsilon_{it} \tag{2}$$

where I_{it} represents the income, B_{it} denotes the non-income (non-pecuniary) benefits of residing in the origin province, such as family ties, social networks, or attachment to place, and C_{it} indicates the costs associated with residing in the origin province. Finally, ϵ_{it} represents the unobserved (idiosyncratic) component of utility, capturing individual-specific preferences and factors not directly measurable within the model.

Next, I define the components I_{it}, B_{it}, and C_{it} of U_{it}:

[9] Unlike Reuveny and W. H. Moore (2009), I do not incorporate interest or discount rates into the utility function in order to keep the model tractable and focus on the contemporaneous effects of climatic and socioeconomic factors on migration decisions.

Income in the origin province is given by:

$$I_{it} = W_{it} + PI_{it} \tag{3}$$

where W_{it} denotes wealth, including assets such as houses, and PI_{it} represents permanent income, such as salary or wages.[10] While Beine, Docquier, and Özden (2011) models income using wages, I define income as the sum of wealth and permanent income.

Benefits associated with remaining in the origin location are represented by B_{it}, which captures non-pecuniary factors such as family ties, social networks, habits, and attachment to place. Following Rapoport, Sardoschau, and Silve (2022), I assume that individuals derive utility from maintaining existing social and cultural connections. Consequently, stronger attachment to the origin province increases the utility of remaining rather than migrating.

Costs in the origin province are given by:

$$C_{it} = DC_{it} + RB_{it} + CCC_{it} \tag{4}$$

where DC_{it} denotes daily costs, which are the costs for living, RB_{it} represents refugee burden, and CCC_{it} denotes climate change costs. Following Beine and C. R. Parsons (2017), I model climate change as a non-pecuniary cost of remaining in the origin province. Given the focus of this paper on the impact of climate change, CCC_{it} can be further defined as:

$$CCC_{it} = T_{it} - P_{it} \tag{5}$$

where T_{it} represents temperature and P_{it} denotes precipitation in origin province i at time t. Higher values of T_{it} are assumed to increase climate-related costs, particularly when temperatures become unusually high. Conversely, higher values of P_{it} are assumed to reduce climate-related costs by mitigating drought conditions and supporting agricultural activity.

[10] The decomposition of income into wealth and permanent income follows the Permanent Income Hypothesis of Friedman (1957). In this study, I assume that individuals who decide to migrate own a single house as part of their wealth and do not sell or rent it because of family heritage.

Thus, I can rewrite the utility function for the origin as in eq:Moriginutilitydetailed:

$$U_{it} = W_{it} + PI_{it} + B_{it} - DC_{it} - RB_{it} - T_{it} + P_{it} \tag{6}$$

According to eq:Moriginutilitydetailed, the utility of a person in the origin city generally depends on wealth, personal income,non-pecuniary benefits associated with remaining in the origin province, the presence of refugees, and the impact of temperature and precipitation.

In the context of Türkiye, higher temperatures, particularly during the summer months, are expected to increase the cost of remaining in the origin province by reducing agricultural productivity, increasing heat exposure, and worsening living conditions. Previous studies have shown that high temperatures can reduce productivity and encourage migration once critical temperature thresholds are exceeded (Bohra-Mishra, Oppenheimer, and Hsiang, 2014; Burzyński et al., 2022; Burke, Hsiang, and Miguel, 2015). Conversely, higher precipitation is assumed to reduce climate-related costs by mitigating drought conditions and supporting agricultural activity (Barrios, Bertinelli, and Strobl, 2006). I also incorporate refugee burden into the cost function because large refugee populations may place additional pressure on local labor markets and public services, thereby increasing the costs associated with remaining in the origin province.

Similarly, the expected utility associated with the destination province depends on a combination of economic, climatic, and migration-related factors. Expected utility in destination province j at time t, U^e_{jt}, is influenced by expected income, favorable climate conditions, migration costs between the origin and destination, daily living costs in the destination, and other factors that increase the attractiveness of the destination, such as improved economic opportunities and future prospects.

First, I express the utility function for the destination province as expected values:

$$U^e_{jt}(I_{jt}, B_{jt}, C_{jt}) = I^e_{jt} + B^e_{jt} - C^e_{jt} + \epsilon_{jt} \tag{7}$$

where ϵ_{jt} represents the unobserved (idiosyncratic) component of utility, capturing individual-specific preferences and factors not directly measurable within the model.

Next, I define the components of U^e_{jt}:

Income in the destination province is given by:

$$I^e_{jt} = PI^e_{jt} \tag{8}$$

where PI^e_{jt} denotes expected personal income in destination province (j) at time (t). Unlike eq:Moriginutilitydetailed and eq:Mdestinationwealt does not explicitly include a wealth component. I focus on expected income because migration decisions are assumed to depend primarily on expected earnings opportunities in the destination location.[11]

Benefits in the destination province are defined as:

$$B^e_{jt} = BCC^e_{jt} + BH^e_{jt} \tag{9}$$

where BCC^e_{jt} represents better climate conditions in destination province j at time t, BH^e_{jt}epresents the expected quality-of-life benefits associated with residing in the destination province, following Rappaport (2009). Individual risk preferences also influence expected destination utility. Individuals who are more willing to accept uncertainty are expected to perceive greater utility from migration than more risk-averse individuals (Becker, 1962).

Costs in the destination province are expressed as:

$$C^e_{jt} = MC^e_{ij} + D_{ij} + DC^e_{jt} \tag{10}$$

I interpret D_{ij} as a measure of geographic separation, whereas MC^e_{ij} captures the broader economic and social costs of migration, including transportation expenses, information costs, and the costs of maintaining social and economic ties across locations.

[11]This simplification follows a common practice in migration models that emphasizes labor-market incentives as the primary determinant of destination choice. While wealth may influence an individual's ability to migrate, incorporating wealth accumulation and asset transfers would substantially increase the model's complexity without altering its central focus on expected economic opportunities and climate-related migration incentives.

where MC^e_{ij} denotes the migration cost associated with moving from origin province (i) to destination province (j), D_{ij} represents the geographic distance between the two locations, and DC^e_{jt} denotes the daily cost of living in destination province (j) at time (t). I interpret the MC^e_{ij} captures the broader economic and social costs of migration, including transportation expenses, information costs, and the costs of maintaining social and economic ties across locations. While gravity models of international trade often interpret distance through iceberg trade costs Anderson and Wincoop (2003) and Santana-Gallego and Paniagua (2022), I use D_{ij} more generally as a measure of the costs and frictions associated with migration.

Combining the expected utility components, the utility function for the destination province can be written as:

$$U^e_{jt} = PI^e_{jt} + BCC^e_{jt} + BH^e_{jt} - MC^e_{ij} - D_{ij} - DC^e_{jt} \tag{11}$$

eq:Mdestinationcomponents indicates that the expected utility of migrating depends on expected income in the destination province, favorable climate conditions, expected quality-of-life benefits, migration costs, geographic distance, and daily living costs.

An individual chooses to migrate if the expected utility of the destination exceeds the utility of remaining in the origin:

$$W_{it} + PI_{it} + B_{it} - DC_{it} - RB_{it} - T_{it} + P_{it} < PI^e_{jt} + BCC^e_{jt} + BH^e_{jt} \quad - MC^e_{ij} - D_{ij} - DC^e_{jt} \tag{12}$$

For simplicity, I assume that expected personal income equals personal income $(PI_{it} = PI^e_{jt})$, expected daily living costs equal daily living costs $(DC_{it} = DC^e_{jt})$, and expected non-pecuniary benefits equal current non-pecuniary benefits $(B_{it} = BH^e_{jt})$.

The migration decision can therefore be simplified as:

$$W_{it} - RB_{it} - T_{it} + P_{it} < BCC^e_{jt} - MC^e_{ij} - D_{ij} \tag{13}$$

eq:Mcombinedallsimplified summarizes the individual's migration decision by comparing the utility of remaining in the origin province with the expected utility of migrating to the destination province. Migration occurs when the expected utility associated with the destination exceeds the utility of remaining in the origin province.

According to eq:Mcombinedallsimplified, migration decisions are determined by comparing the utility of remaining in the origin province with the expected utility of migrating to the destination province. Higher temperatures increase the costs associated with remaining in the origin province, thereby increasing the incentive to migrate. Temperature is not included as a cost in the destination province because individuals are assumed to migrate in search of more favorable climate conditions, consistent with Rappaport (2009). I also include refugee burden as a component of the cost function because larger refugee populations may place additional pressure on local labor markets, public services, and social integration, thereby increasing the costs associated with remaining in the origin province.

The probability of migration, defined as the probability that $U^e_{jt} > U_{it}$, can be expressed using an exponential function:

$$P(U^e_{jt} > U_{it}) = \min\left(1, \frac{e^{U^e_{jt}}}{e^{U_{it}}}\right) \tag{14}$$

Substituting the utility functions, I get:

$$P(U^e_{jt} > U_{it}) = \min\left(1, \frac{e^{BCC^e_{jt} - TMC^e_{ij} - D_{ij}}}{e^{W_{it} - RB_{it} - T_{it} + P_{it}}}\right) \tag{15}$$

I next consider the probability of choosing destination j from the set of all possible destinations. Assuming there are multiple possible destinations, the individual will choose the destination that maximizes his utility. The probability of choosing destination j out of all possible destinations is given by:

$$P(\text{Choose } j \mid \text{Migration}) = \frac{e^{U^e_{jt}}}{\sum_{k \in J} e^{U^e_{kt}}} \tag{16}$$

Combining eq:Mprobabilityuj¿ui and eq:Mprobabilitycitychoose yields eq:Mprobabilitycombined, which gives the overall probability of migrating from origin i to destination j:

$$P(\text{Migration to } j) = \min\left(1, \frac{e^{BCC^e_{jt} - MC^e_{ij} - D_{ij}}}{e^{W_{it} - RB_{it} - T_{it} + P_{it}}}\right) \cdot \left(\frac{e^{BCC^e_{jt} - MC^e_{ij} - D_{ij}}}{\sum_{k \in J} e^{BCC^e_{kt} - MC^e_{ik} - D^e_{ik}}}\right) \tag{17}$$

eq:Mprobabilitycombined shows that an individual's migration decision depends on two components. First, the expected utility associated with the destination province must exceed the utility of remaining in the origin province. Second, among all potential destination provinces, the individual chooses the destination that provides the highest expected utility. Consequently, migration occurs toward the destination that maximizes expected utility.

According to eq:Mprobabilitycombined, the probability of migration from origin province i to destination province j depends on climatic, economic, and geographic factors. Higher temperatures, greater refugee burden, longer migration distance, and higher migration costs reduce the attractiveness of migration, whereas higher precipitation and more favorable destination conditions increase the likelihood of migration. These relationships provide the theoretical foundation for the empirical analysis presented in the following section.

Thus, the migration equation can be expressed as:

$$M_{ijt} = \frac{e^{BCC^e_{jt} - MC^e_{ij} - D_{ij}}}{e^{W_{it} - RB_{it} - T_{it} + P_{it}}} \cdot \eta_{ijt} \tag{18}$$

where η_{ijt} is the unobserved (idiosyncratic) error term, capturing individual-specific preferences and other factors not directly observed in the model.

4.2 Gravity Model

The gravity model aims to formalize and predict the geography of flows or interactions. Originally developed to analyze international trade, the gravity model posits that flows between two units are proportional to the product of their economic sizes (typically GDP) and inversely proportional to the distance between them (Anderson and Wincoop, 2003; Anderson and Yotov, 2025). While first applied to trade, it has more recently been extended to migration flows (Afifi and Warner, 2008; Beine, Docquier, and Özden, 2011; Beine and C. R. Parsons, 2017; Grohmann, 2023).

The basic gravity equation, originally introduced by Tinbergen (1962) and subsequently developed by McCallum (1995), Anderson and Wincoop (2003), Santos Silva and Tenreyro (2006), and Anderson and Yotov (2025), can be expressed as:

$$T_{ij} = \alpha \frac{Y_i Y_j}{D_{ij}} \tag{19}$$

where T_{ij} is the trade flow from country i to country j, Y_i and Y_j are the GDPs of countries i and j, respectively, D_{ij} is the distance between the two countries, and α is a constant.

Santos Silva and Tenreyro (2006) addressed the problem of zero trade flows, which can lead to biased estimates in log-linearized gravity models. They proposed estimating the gravity equation using the PPML estimator, which naturally accommodates zero trade flows and is robust to heteroskedasticity. The gravity model can then be written as:

$$E[T_{ij}] = \exp(\beta_0 + \beta_1 \ln Y_i + \beta_2 \ln Y_j + \beta_3 \ln D_{ij} + X'_{ij}\beta) \tag{20}$$

where X_{ij} represents additional independent control variables.

Rapoport, Sardoschau, and Silve (2022) and Santana-Gallego and Paniagua (2022) extended the gravity model by incorporating additional explanatory variables that influence bilateral flows. The generalized gravity model can be written as:

$$T_{ij} = \alpha \left(\frac{Y_i^{\beta_1} Y_j^{\beta_2}}{D_{ij}^{\beta_3}} \right) \exp(\gamma Z_{ij} + u_{ij}) \tag{21}$$

where Z_{ij} represents additional explanatory variables affecting bilateral flows, and u_{ij} is an error term.

In the context of migration, the gravity model can be adapted by replacing trade flows with migration flows and incorporating economic and climatic variables as the main determinants. Thus, the gravity equation becomes:

$$M_{ij} = \alpha\, GDP_i^{\beta_1}\, GDP_j^{\beta_2}\, D_{ij}^{\beta_3} \times \exp(\gamma_1 Temp_i + \gamma_2 Temp_j + \gamma_3 Prec_i + \gamma_4 Prec_j + \gamma_5 RB_i)\, \epsilon_{ij} \tag{22}$$

where M_{ij} is the migration flow from origin i to destination j, GDP_i and GDP_j measure the economic size of the origin and destination, D_{ij} is the distance between them, $Temp_i$ and $Temp_j$ denote average temperatures, $Prec_i$ and $Prec_j$ denote precipitation levels, and RB_i represents the refugee burden in the origin province. The error term ϵ_{ij} captures unobserved factors affecting migration flows.

4.3 Econometric Specification

To empirically examine the determinants of internal migration in Türkiye, I translate the theoretical framework developed in the previous sections into an econometric specification. The empirical strategy follows a gravity-type modeling approach that is grounded in the RUM and widely used in migration studies. This framework explains migration flows between origin and destination provinces as a function of climatic conditions, economic characteristics, spatial frictions, and nonlinear temperature effects.

I model migration flows using a gravity-based migration equation that incorporates climatic, economic, and geographic determinants. To capture potential nonlinear responses to climate conditions, I allow the effect of temperature to vary nonlinearly, reflecting the possibility that migration responses become stronger under more severe climatic conditions.

4.3.1 Gravity-Based Migration Model

The empirical specification combines insights from the RUM and the gravity framework to analyze the determinants of internal migration flows. This integrated approach links individual migration decisions, based on utility maximization, with the broader structural factors that shape migration flows between origin and destination provinces.

RUM provides the microeconomic foundation for migration behavior, suggesting that individuals choose to migrate when the expected utility in a destination location exceeds the utility of remaining in the origin location. These decisions are influenced by economic opportunities, environmental conditions, and migration costs (Beine and C. Parsons, 2015; Beine and C. R. Parsons, 2017; Ruiz, 2017).

The gravity model complements this framework by capturing the spatial structure of migration flows. Originally developed to analyze international trade by Tinbergen (1962) and later refined by Anderson and Wincoop (2003), the gravity model predicts that interactions between locations increase with economic size and decrease with distance. This framework has been widely applied to migration studies (Beine and C. Parsons, 2015; Backhaus, Martinez-Zarzoso, and Muris, 2015; Beine and C. R. Parsons, 2017).

Combining these two frameworks provides a natural way to model migration flows as a function of climatic conditions, economic characteristics, and spatial frictions. Following the empirical approaches of Backhaus, Martinez-Zarzoso, and Muris (2015), Beine and C. R. Parsons (2017), and Ruiz (2017), I specify the following gravity-based migration model, allowing for nonlinear temperature effects:

$$\ln(M_{ijt}) = \alpha + \beta_1 \ln(T_{it}) + \beta_2 \ln(T_{it})^2 + \beta_3 \ln(T_{jt}) + \beta_4 \ln(T_{jt})^2 + \beta_5 \ln(P_{it}) + \beta_6 \ln(P_{jt}) + \beta_7 \ln(GDP_{it}) + \beta_8 \ln(GD \tag{23}$$

where M_{ijt} denotes the migration flow from origin province i to destination province j in year t. The variables T_{it} and T_{jt} represent temperature in the origin and destination provinces, respectively, and enter the model in both linear and quadratic forms to capture potential nonlinear effects. The variables P_{it} and P_{jt} denote precipitation in the origin and destination provinces, GDP_{it} and GDP_{jt} represent economic conditions in the origin and destination provinces, D_{ij} is the geographic distance between the two provinces and proxies migration costs, and RB_{it} denotes the refugee burden in the origin province.[12]

The model includes origin, destination, and year fixed effects to control for unobserved heterogeneity. The term λ_i captures time-invariant characteristics of origin provinces, θ_j captures time-invariant characteristics of destination provinces, and δ_t denotes year fixed effects that control for common shocks affecting migration patterns over time. The error term ϵ_{ijt} captures unobserved factors affecting migration flows. In the empirical analysis, robust standard errors are clustered at the origin–destination pair level to account for serial correlation and heteroskedasticity within province pairs.

4.3.2 Nonlinear Climate Effects

Although the gravity model can be specified using linear climate variables, a growing body of literature suggests that the effects of climatic conditions on migration are often nonlinear. In particular, moderate changes in temperature may have limited effects on migration, whereas more extreme temperatures can generate substantially greater migration responses (Bohra-Mishra, Oppenheimer, and Hsiang, 2014; Burke, Hsiang, and Miguel, 2015; Hsiao, 2026).

To capture these potential nonlinear effects, I extend the model by including quadratic terms for temperature. The nonlinear specification is given by:

$$\ln(M_{ijt}) = \beta_1 \ln(T_{it}) + \beta_2 (\ln T_{it})^2 + \beta_3 \ln(T_{jt}) + \beta_4 (\ln T_{jt})^2 + \beta_5 \ln(P_{it}) + \beta_6 \ln(P_{jt}) + \theta \ln(D_{ij}) + \sum \gamma Z_{it} + \lambda_i + \theta_j + \tag{24}$$

In this specification, the quadratic temperature terms allow the model to capture nonlinear responses of migration to climatic conditions. These terms enable the identification of threshold effects, whereby migration responses may differ across the temperature distribution.

[12] Equation eq:MestimationmodelLn is presented in log-linear form for expositional clarity. In the empirical analysis, the three-way fixed effects and Hausman–Taylor models are estimated using the logarithm of migration flows as the dependent variable, whereas the PPML specification is estimated using migration flows in levels. This follows Santos Silva and Tenreyro (2006), who show that PPML provides consistent estimates for gravity-type models in the presence of heteroskedasticity and zero flows. The explanatory variables may still enter the PPML specification in logarithmic form.

The vector Z_{it} includes additional time-varying control variables measured at the origin province level, such as wildfire activity, electricity consumption, and the refugee burden, depending on the specification. The model also includes origin, destination, and year fixed effects to control for unobserved spatial heterogeneity and common macroeconomic shocks affecting migration patterns over time.

4.4 Estimation Methods

To estimate the gravity-based migration model, I employ three complementary estimation approaches. First, I estimate the model using Three-Way Fixed Effects to control for unobserved heterogeneity across origin provinces, destination provinces, and time periods. Second, I apply the PPML estimator, which has become the standard approach for estimating gravity-type models because it accounts for zero flows and heteroskedasticity. Finally, I employ the Hausman-Taylor estimator to address potential endogeneity while allowing the estimation of time-invariant variables such as geographic distance.

Together, these estimation approaches allow me to assess the robustness of the results and ensure that the estimated relationships between climatic variables and migration flows are not driven by model-specific assumptions.

4.4.1 Three-Way Fixed Effects

I begin by estimating the model using a Three-Way Fixed Effects specification. This approach controls for unobserved heterogeneity across origin and destination provinces and across time periods by including origin, destination, and year fixed effects in the regression.

Including these fixed effects allows me to control for time-invariant characteristics of provinces that may influence migration decisions, such as geographic conditions, long-term economic structure, or cultural factors. Year fixed effects capture common macroeconomic or national shocks affecting migration patterns over time. This specification is widely used in migration studies to account for unobserved spatial heterogeneity and improve the consistency of estimated coefficients.

4.4.2 PPML Method

I also estimate the gravity model using the PPML estimator. Santos Silva and Tenreyro (2006) show that PPML provides consistent estimates for gravity-type models even when the dependent variable contains zeros, a common feature of migration and trade data. Unlike log-linearized ordinary least squares estimations, PPML estimates the model using the dependent variable in levels. This approach avoids biases arising from heteroskedasticity and allows zero migration flows to be included in the estimation (Linders and Groot, 2006; Santos Silva and Tenreyro, 2011).

As a result, PPML has become the preferred estimator for gravity models in both trade and migration studies. The method has been widely applied in migration research (Beine, Bertoli, and Fernández-Huertas Moraga, 2016; Beine and C. R. Parsons, 2017). For example, Ruiz (2017) applied PPML to study the effect of climatic shocks on internal migration in Mexico, demonstrating its suitability for analyzing migration flows.

4.4.3 Hausman–Taylor Estimation

In this study, I apply the HT estimator to model internal migration flows, allowing the estimation of time-invariant variables such as geographic distance while addressing potential endogeneity. I also apply the Hausman-Taylor estimator to address potential endogeneity and unobserved heterogeneity in the migration model. Traditional fixed-effects models control for unobserved heterogeneity but cannot estimate coefficients for time-invariant variables, such as geographic distance. Random effects models allow the inclusion of time-invariant regressors but yield biased estimates when regressors are correlated with individual-specific effects. The Hausman-Taylor estimator overcomes these limitations by using internal instruments to obtain consistent estimates when selected regressors are correlated with individual-specific effects (Hausman and Taylor, 1981; Wooldridge, 2010). It provides consistent estimates for both time-varying and time-invariant variables by exploiting variation within and between provinces, making it well-suited for migration studies in which variables such as distance remain constant while other characteristics vary over time.

Empirical applications confirm its usefulness. Egger and Pfaffermayr (2004) applied the Hausman-

Taylor estimator to address endogeneity in trade and foreign direct investment analyses, while Serlenga and Shin (2004) extended the framework to account for unobserved time-specific effects. Similarly, Baltagi, Bresson, and Pirotte (2003) demonstrated that the HT estimator exploits the within- and between-variation of strictly exogenous variables as internal instruments, allowing consistent estimation when regressors are correlated with individual-specific effects.

5 Data

This section describes the data used to analyze internal migration flows in Türkiye. I first describe the construction of the origin–destination panel and then summarize the key climatic, economic, and demographic variables used in the analysis.

Administratively, Türkiye is divided into 81 provinces (*iller*), which serve as the unit of analysis in this study.[13] Using these administrative units, I construct an origin–destination panel that includes migration flows between all province pairs over the sample period. In addition to the full sample covering all 81 provinces, I also consider two subsamples: 30 provinces most exposed to climatic risks and 17 provinces identified as drought-affected, which are examined in complementary analyses.

Climate-related factors play a central role in constructing the subsamples used in the analysis. In particular, I classify provinces according to their exposure to climatic stressors, including droughts, extreme heat, wildfires, and floods. Provinces that exceed predefined thresholds for these indicators, including measures based on drought indices, are classified as climate-exposed. Using this approach, I identify a climate-exposed sample of 30 provinces and a narrower sample of 17 drought-affected provinces. These subsamples complement the full-sample analysis by allowing me to examine whether migration responses are stronger in areas facing greater climatic stress.

The dataset covers the period from 2008 to 2024. Using data for all 81 provinces, I construct a balanced origin–destination–year panel of internal migration flows. Excluding self-migration, each

[13]In Türkiye, the administrative unit officially called a “province” (*il*) is often translated into English as a “city.” Each province contains multiple districts, towns, and villages; however, in statistical and economic analyses, the province is treated as the unit of observation. Accordingly, when we refer to “cities” in this study, we mean provinces at the Nomenclature of Territorial Units for Statistics-3 (NUTS-3) level.

year includes $81 \times 80 = 6{,}480$ ordered province pairs, yielding a total of $6{,}480 \times 17 = 110{,}160$ observations. The climate-exposed subsamples are derived from this full panel and therefore contain fewer observations.

To illustrate the structure of the migration framework, consider a simplified example with two origin cities, labeled A and B, and five destination cities, labeled A, B, C, D, and E, as shown in Figure 2. In this setting, each origin city can send migrants to all destination cities except itself, reflecting the restriction against self-migration. For example, city A can send migrants to cities B, C, D, and E, but not to itself, while city B can send migrants to cities A, C, D, and E. This simplified illustration generalizes to the full dataset, where all 81 provinces are treated symmetrically. In each year, every province can send migrants to the remaining 80 provinces, resulting in a complete set of ordered origin–destination pairs, excluding self-migration.

In the full sample, migration flows are modeled as unidirectional movements from origin to destination, with no restrictions imposed on migration between specific provinces beyond the exclusion of self-migration. This general structure is maintained when focusing on climate-related subsamples, including the 30 most climate-exposed provinces and the 17 drought-affected provinces. While migration patterns in these subsamples more closely reflect south-to-north and less-developed-to-more-developed movements commonly discussed in the climate migration literature, migration flows are still allowed between all province pairs within the relevant sample, ensuring consistency between the full-sample and subsample analyses.[14]

[14]Beine and C. Parsons, 2015 document differences in migration determinants between movements toward the global North and South, highlighting the role of economic incentives, skill composition, and distance.

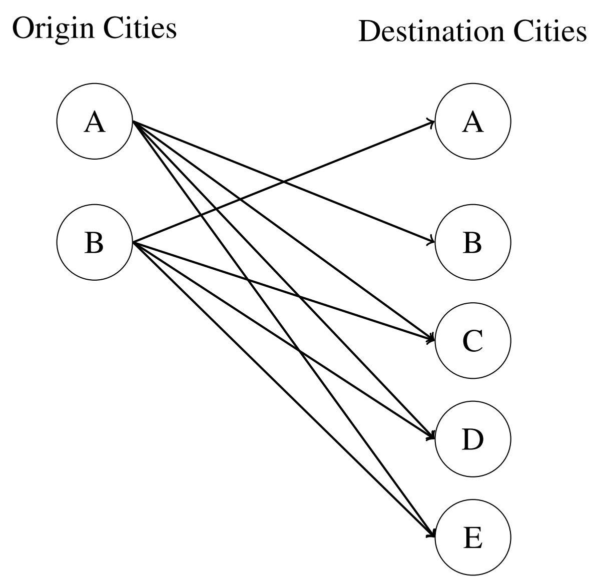

Figure 2: Illustration of Origin–Destination Migration Links

Note: The figure presents a simplified example with two origin cities (A, B) and five potential destination cities (A, B, C, D, E). Arrows indicate allowed origin-destination flows under the restriction against self-migration. The same structure applies in the empirical analysis, where each province can send migrants to all other provinces in a given year.

Figure 3 illustrates the spatial distribution of major climatic stressors across provinces in Türkiye and provides the basis for constructing the climate-related subsamples used in the analysis. The map highlights substantial regional heterogeneity in exposure to droughts, floods, and wildfires. Drought exposure is widespread across much of the country, flood risk is more pronounced in northern provinces, and wildfire activity is concentrated mainly in the south. These spatial patterns reflect regional differences in geography and climatic conditions documented in the literature (Burzyński et al., 2022). Although the main empirical analysis includes all 81 provinces, the figure identifies the provinces most exposed to climatic stress. Specifically, the climate-exposed sample comprises 30 provinces with relatively high exposure to these stressors, while a subset of 17 provinces is classified as drought-affected. These subsamples are used in complementary analyses to examine whether migration responses are stronger in areas facing more severe climatic stress, while preserving the full-sample migration framework.

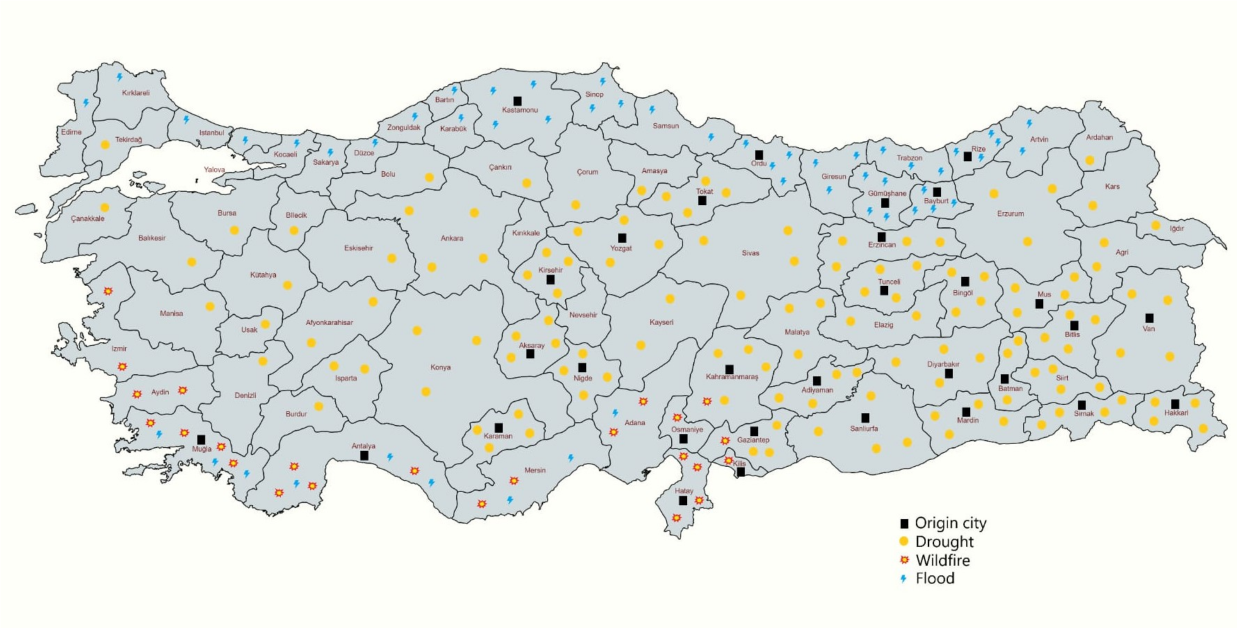

Figure 3: Spatial Distribution of Climatic Exposure across Provinces in Türkiye

Note: The figure illustrates the spatial distribution of major climatic stressors across provinces in Türkiye. Provinces affected by drought conditions are identified using the Bagnouls–Gaussen aridity index and are marked in yellow. Provinces primarily exposed to floods are indicated in blue, while provinces experiencing significant wildfire activity are shown in red with yellow interiors. Black squares denote provinces included in the climate-exposed subsamples used in the analysis. Drought exposure is widespread across the country, flood exposure is more prevalent in northern regions, and wildfire activity is concentrated mainly in southern provinces. *Data source:* Turkish State Meteorological Service (TSMS)

Table 1 reports summary statistics for all variables used in the analysis, including the number of observations, means, standard deviations, and ranges. Most variables are observed for the full sample period, yielding 110,160 observations. However, several origin-level socioeconomic variables have fewer observations due to differences in data availability over time. In particular, information on Syrian refugees and crop production values is available only for a shorter subperiod, while data on fertility rates, hospitals, students, and students per classroom are reported only for selected years.

Table 1: Summary Statistics

	Obs.	Mean	Std. Dev.	Min	Max
Migration flow (origin to destination)	110,160	406	1,208	0	30,036
Distance (km)	110,160	762	395	30	2,016
Average temperature (°C)	110,160	14.6	3.3	4.1	22.1
Mean maximum temperature (°C)	110,160	20.2	3.0	10.5	28.1
Mean minimum temperature (°C)	110,160	9.1	3.9	-2.8	18.6
Winter minimum temperature (°C)	110,160	0.6	5.0	-20.5	10.9
Summer maximum temperature (°C)	110,160	31.1	3.4	21.1	40.1
Total precipitation (mm)	110,160	643	322	121	3,097
Winter precipitation (mm)	110,160	229	158	6	1,083
Summer precipitation (mm)	110,160	87	91	0	868
GDP (million TL)	110,160	91,518.58	486,800.00	568.51	13,000,000
GDP pc ($)	110,160	8,195	3,498	2,713	24,452
GDP pc (TL)	110,160	65,239	104,615	3,888	802,669
Electricity consumption pc (kWh)	110,160	2,573	1,670	443	11,004
Divorces	110,160	1,724	3,733	30	35,338
Crop production value	90,720	1,645,574	2,194,976	13,650	19,592,152
Fertility rate	103,680	2.0	0.7	1.1	4.7
Hospitals	103,680	18.5	26.8	1	238
Students	84,240	66,985	115,474	2,695	990,111
Literacy rate (%)	110,160	95.0	3.3	78.7	99.1
Population	110,160	981,593	1,760,206	74,412	15,907,951
Population density	108,880	95.0	114.1	10.1	2,808
Students per classroom	84,240	21.0	5.6	12	48
Cereal production	110,160	1,367,386	1,749,240	335	19,253,855
Number of wildfires	77,760	34	50	0	322
Wildfire-affected area (ha)	77,760	281	2,589	0	60,367
Foreign residents	110,160	7,922	43,100	24	675,678
Syrian refugees	90,720	32,730	86,826	0	655,505

Note: The table reports summary statistics for all variables used in the analysis. Most variables are observed for the full sample period, resulting in 110,160 observations. However, several origin-level socioeconomic variables have fewer observations due to limited temporal coverage. In particular, data on Syrian refugees and crop production

Distance data were taken from the Republic of Türkiye's General Directorate of Highways (GDH).[15] Distances are measured along the road network rather than as straight-line (Euclidean) distances and are treated as time-invariant. Distances between provinces in the dataset range from about 30 kilometers (km) to just over 2,000 km.

Migration data were obtained from the TSI, Population Statistics Portal.[16] Migration flows are reported annually at the province level and record movements from origin to destination provinces. During data preparation, province names were carefully standardized to avoid sorting inconsistencies caused by Turkish characters. Zero migration flows between some origin–destination pairs are observed in the data and are retained in the analysis. Migration flows range from zero to approximately 30,000 individuals.

Monthly temperature and precipitation data were obtained from the TSMS.[17] ”Using these monthly records, I construct annual climate variables by calculating yearly averages for temperature and annual totals for precipitation. In addition to these aggregate measures, I also construct seasonal indicators to capture temperature extremes, including summer maximum and winter minimum temperatures. All climate variables were constructed separately for origin and destination provinces, allowing for distinct push and pull effects of climatic conditions in migration decisions. Across provinces and years, average annual temperatures range from approximately 4°C in colder regions to about 22°C in warmer areas, while annual precipitation ranges from roughly 120 mm to 3,100 mm. These seasonal measures reveal substantial variation in both summer heat and winter cold.

Population data for origin provinces were collected from TSI.[18] Within the sample, provincial populations range from approximately 75,000 in the smallest provinces to nearly 16 million in Istanbul, the country's most populous city.

Economic variables were obtained from the TSI. Provincial GDP pc, measured in U.S. dollars, ranges from approximately \$2,700 to over \$24,000 across provinces and years. Total provincial

[15]Data Source: `https://www.kgm.gov.tr/`
[16]Data source: `https://nip.tuik.gov.tr`
[17]Data source: `https://www.mgm.gov.tr/eng`
[18]Data Source: `https://data.tuik.gov.tr`

GDP is also reported in Turkish lira. Electricity consumption is measured in kilowatt-hours pc (kWh) and varies from fewer than 500 to over 11,000 kWh. Additional socioeconomic indicators include agricultural production values, fertility rates, hospitals, students, literacy rates, population levels, and population density, all reported at the provincial level by TSI.

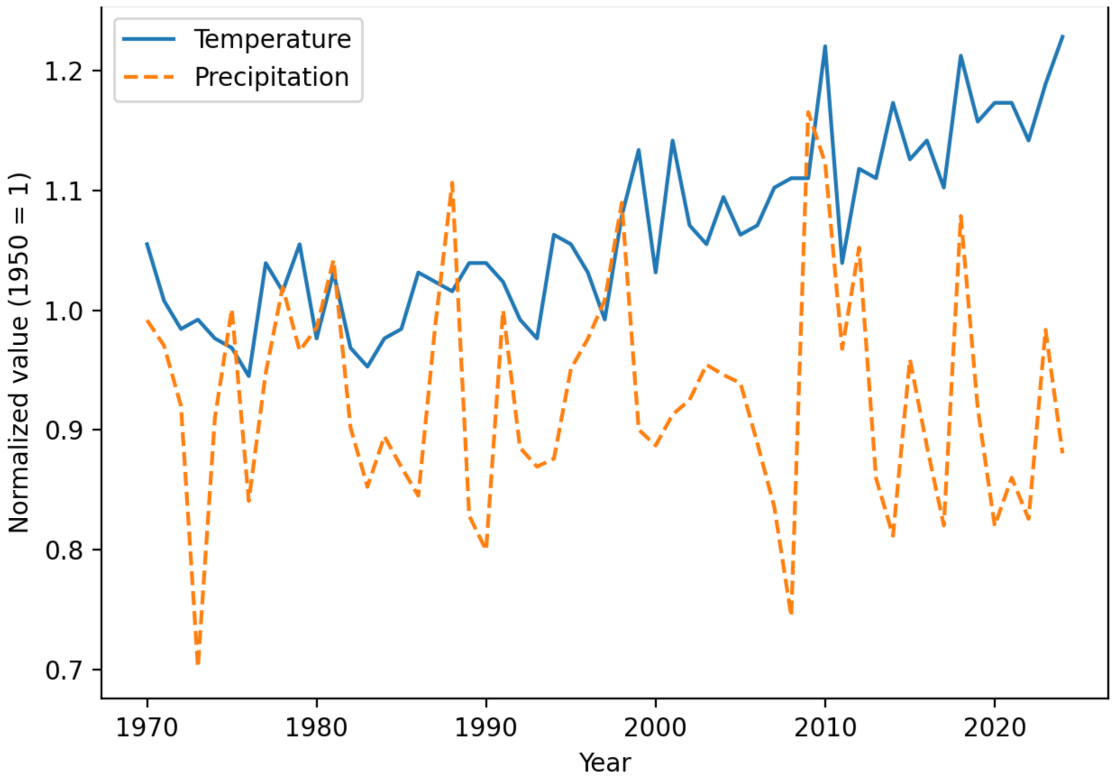

Figure 4: Normalized Temperature and Precipitation in Türkiye, 1970–2024

Note: Temperature and precipitation are normalized to their 1950 values (1950 = 1) to facilitate comparison across variables measured in different units. Using 1950 as the reference year provides a common baseline before the pronounced warming observed in later decades.

Environmental stress indicators include wildfire data obtained from the General Directorate of Forestry in Türkiye (GDF).[19] These data record both the annual number of wildfire incidents and the total wildfire-affected area, measured in hectares. Across provinces and years, the number of wildfires ranges from 0 to over 300, while the affected area ranges from 0 to approximately 60,000 hectares.

[19]Data source: https://www.ogm.gov.tr/

Information on refugees was obtained from the DGMM.[20] Provincial counts of Syrian refugees are included to capture potential migration pressures associated with large-scale refugee inflows following the Syrian conflict. I also construct a binary indicator equal to one from 2014 onward, corresponding to the establishment of the Directorate General of Migration Management (DGMM).[21]

Figure 4 illustrates the evolution of average temperature and precipitation in Türkiye, with both series normalized to their 1950 values. The figure shows a clear upward trend in average temperature accompanied by a gradual decline in precipitation. These patterns are consistent with previous studies (Beine and C. R. Parsons, 2017; Berlemann and Steinhardt, 2017). The figure also indicates greater variability in both series over time, suggesting increasingly pronounced climatic fluctuations in recent decades. Longer-term climate trends are presented in **??**, Figure A.2 and Figure A.3.

6 Identification Strategy

Identification relies on within-province variation in climatic conditions over time. Origin and destination fixed effects absorb time-invariant provincial characteristics, while year fixed effects account for common national shocks. Consequently, the climate coefficients are identified from how migration flows respond to changes in annual average temperature and precipitation within provinces over time. The empirical specifications include both linear and quadratic temperature terms, allowing migration responses to vary nonlinearly across the temperature distribution and capture potential threshold effects.

The identifying assumption is that these short-run climatic fluctuations are plausibly exogenous to migration decisions; that is, they are not caused by migration and, conditional on the fixed effects, are plausibly uncorrelated with other time-varying determinants of mobility (Dell, Jones, and Olken, 2014). The primary threat to identification is therefore an omitted time-varying factor that is correlated with both climatic conditions and migration. Depending on the specification, I addi-

[20]Data source: `https://en.goc.gov.tr/`

[21]The DGMM was established in 2014 to oversee migration and refugee-related policies in Türkiye.

tionally control for several observable channels that may influence migration decisions, including economic opportunities through origin and destination GDP pc, demographic pressures through the share of Syrian refugees in the origin province, and environmental and infrastructure conditions through wildfire exposure and electricity consumption. Geographic distance captures the spatial frictions emphasized by the gravity framework and, because it is time invariant, is identified only in the Hausman-Taylor specification.

Although this approach substantially reduces concerns about omitted-variable bias, the estimates should be interpreted as conditional associations rather than as strictly causal effects. Nevertheless, by exploiting within-province variation in climatic conditions while controlling for a wide range of economic, demographic, and geographic factors, the analysis provides credible evidence on the relationship between climate variability and internal migration in Türkiye.

7 Empirical Results

This section presents the empirical evidence on the relationship between climate conditions and internal migration in Türkiye over the period 2008–2024. I begin with graphical evidence illustrating the association between temperature conditions and migration patterns, then turn to the regression analysis. The results are organized around three samples: the full sample of 81 provinces, a subsample of 30 climate-exposed provinces, and a subsample of 17 drought-exposed provinces. For each sample, I report results from three complementary estimation approaches: three-way fixed effects, PPML, and Hausman–Taylor. Climatic and socioeconomic controls are introduced sequentially across specifications to evaluate their contribution to migration. The main analysis reports the preferred log–Celsius temperature specification, while the corresponding mean-centered Kelvin specifications are presented as robustness checks in **??**.

7.1 Graphical Analysis of Temperature and Migration

Before presenting the regression results, I first examine the relationship between temperatures and internal migration graphically. These figures summarize the data patterns and provide preliminary evidence supporting the nonlinear temperature specification used in the empirical analysis. I first present the relationship between migration and temperature conditions at the origin and destination provinces, and then report predicted migration rates across discrete temperature categories obtained from the gravity model estimated using PPML.

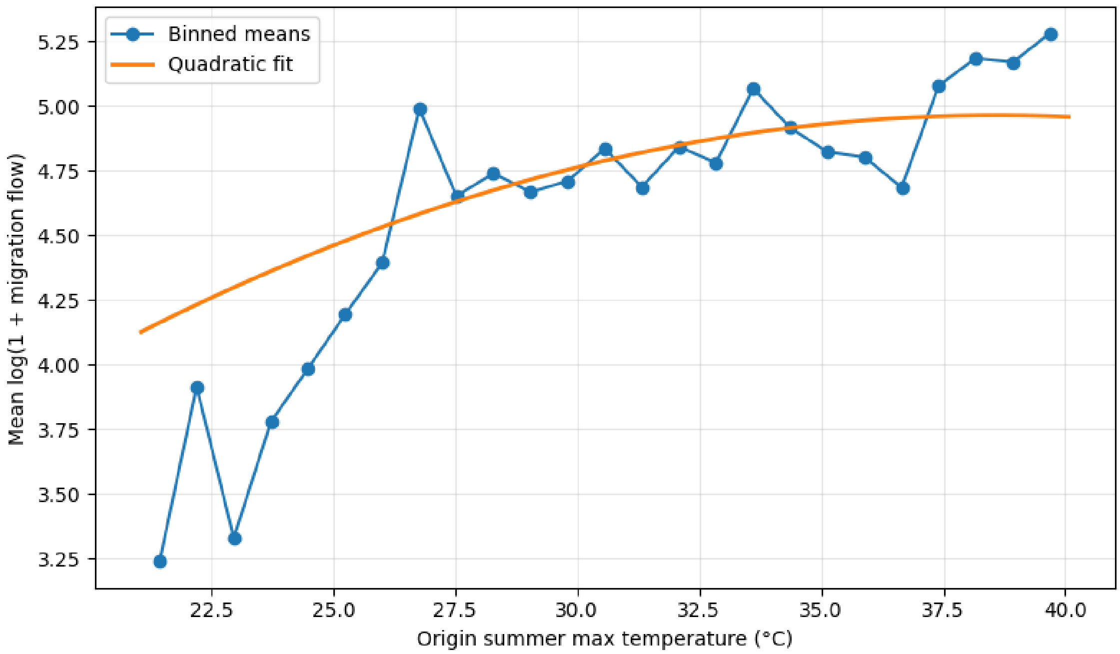

Figure 5: Migration Flows by Origin Summer Maximum Temperature

Note: The figure plots binned averages of $\log(1 + \text{migration flows})$ against origin summer maximum temperature in (°C). The solid curve represents the fitted quadratic relationship.

Figure 5 plots binned averages of migration flows against origin summer maximum temperature. Migration flows initially increase as summer temperatures rise, suggesting that higher temperatures in origin locations may act as a push factor. However, this pattern does not persist across the full temperature range. At higher temperatures, the fitted relationship flattens and then declines slightly, although the binned averages remain variable across the upper temperature range. The fitted quadratic curve highlights this nonlinear relationship.[22]

Figure 6 plots binned averages of migration flows against destination winter minimum temper-

[22]The descriptive figures use seasonal temperature extremes, summer maximum at the origin and winter minimum at the destination, whereas the regression analysis uses average annual temperature. Because seasonal extremes capture only one tail of the temperature distribution, the two measures need not display identical curvature.

ature. Migration flows are relatively low toward destinations with very cold winter temperatures and generally increase as winter minimum temperatures rise. This pattern suggests that migrants are more likely to move toward destinations with milder winter conditions. The fitted quadratic curve highlights this nonlinear relationship.

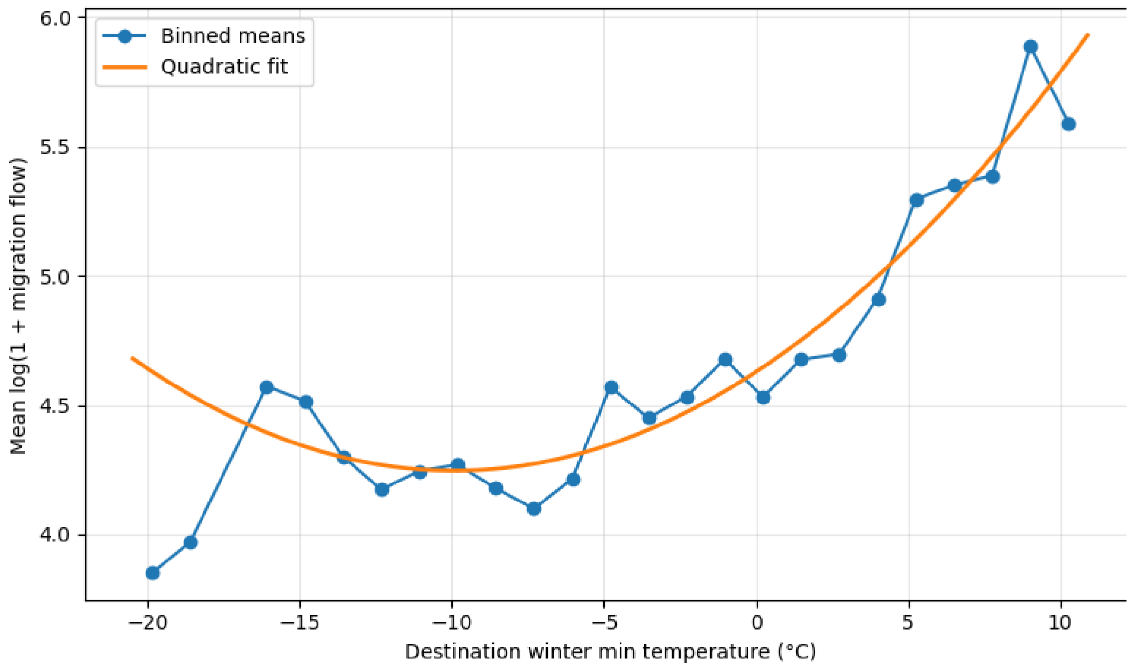

Figure 6: Migration Flows by Destination Winter Minimum Temperature

Note: The figure plots binned averages of $\log(1 + \text{migration flows})$ against destination winter minimum temperature in (°C). The solid curve represents the fitted quadratic relationship.

Figure 7 reports predicted migration rates across destination temperature categories obtained from a PPML gravity model controlling for distance, destination GDP pc, and destination population. Predicted migration rates differ systematically across combinations of origin and destination temperature categories. Migrants from cold-origin provinces exhibit the highest predicted migration rates toward cold and moderate destinations, with migration declining sharply toward hot destinations. In contrast, predicted migration rates for migrants from hot-origin provinces increase steadily as destination temperatures become warmer, reaching their highest level in the hot destination category. Migrants from moderate-origin provinces exhibit predicted migration rates that decline gradually as destination temperatures increase. Notably, the predicted migration rates for hot- and moderate-origin provinces cross between the moderate and hot destination categories. Together, these patterns suggest that the effect of destination climate depends on the origin province's climatic conditions.

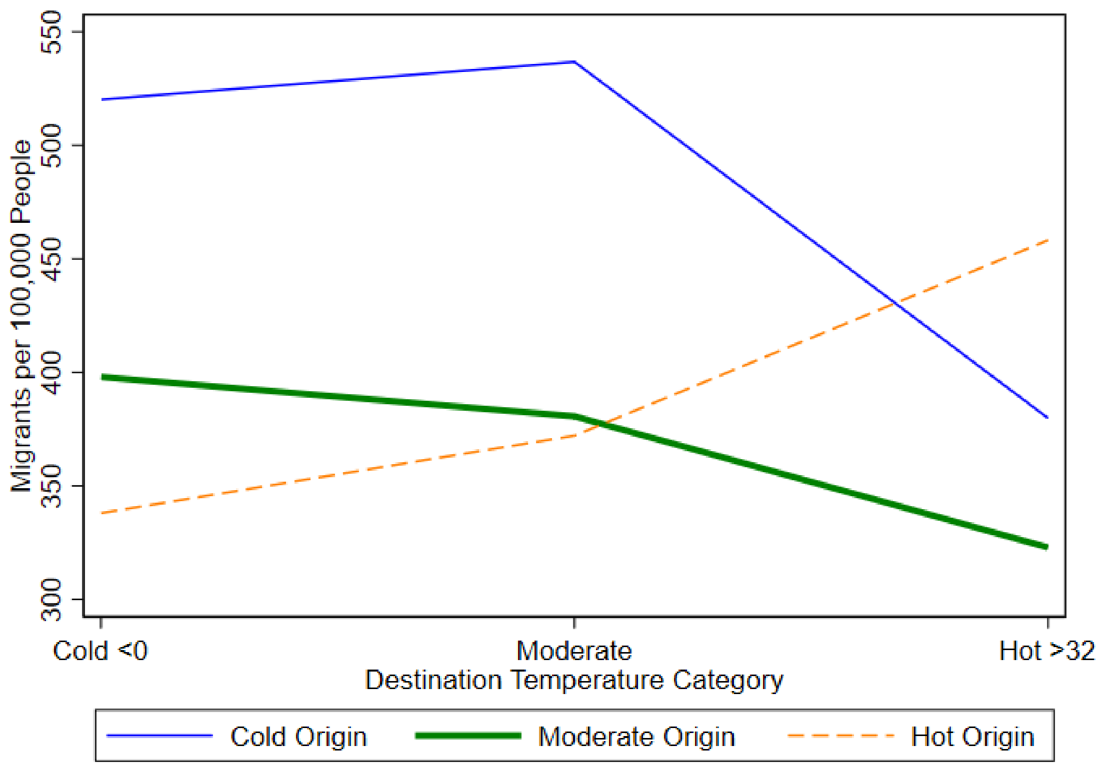

Figure 7: Predicted Migration Rates by Origin and Destination Temperature Categories

Note: The figure reports predicted migration rates (migrants per 100,000 people) across destination temperature categories. Estimates are obtained from the gravity model controlling for distance, destination GDP pc, and destination population. Temperature categories are defined as Cold (< 0°C), Moderate, and Hot (> 32°C). Lines correspond to migrants from cold, moderate, and hot origin regions.

Additional graphical evidence is provided in **??**, where I present Figure A.4–Figure A.13. These figures illustrate alternative temperature–migration relationships using different temperature measures and normalization approaches. The qualitative patterns are consistent with those reported in the main text.

7.2 Full Sample Results

This subsection reports the regression results for the full sample of 81 provinces in Türkiye. The primary specification is a three-way fixed-effects model that controls for origin, destination, and year effects, complemented by PPML and the Hausman–Taylor estimator. These results provide a benchmark for the subsample analyses presented later.

7.2.1 Three-Way Fixed Effects Estimates: Full Sample

Table 2 reports the three-way fixed-effects estimates for the full sample of 81 provinces over the period 2008–2024. The models are estimated sequentially, with climatic and socioeconomic controls introduced progressively. All specifications include origin, destination, and year fixed effects, and standard errors are clustered at the origin-destination pair level.

Model 1 includes only the origin temperature and its quadratic term. The linear coefficient is negative, while the quadratic term is positive, and both are statistically significant. This sign pattern implies a U-shaped relationship between origin temperature and out-migration, with migration initially declining as temperatures rise but increasing once temperatures exceed a threshold. The estimated nonlinear relationship is consistent with temperature extremes at the origin acting as push factors for migration. Model 2 adds precipitation, which enters with a negative and statistically significant coefficient, while the estimated temperature coefficients retain their signs and statistical significance. Model 3 further includes electricity consumption, which is positive and statistically significant, suggesting that provinces with higher levels of economic activity also experience greater migration flows.

Table 2: Three-Way Fixed Effects Estimates: Full Sample of 81 Provinces

	Model 1	Model 2	Model 3	Model 4	Model 5
Origin Temperature (°C)	-0.794^{***}	-0.769^{***}	-0.689^{***}	-0.418^{**}	-0.299^{*}
	(0.156)	(0.158)	(0.157)	(0.167)	(0.167)
Origin Temperature^2	0.203^{***}	0.196^{***}	0.173^{***}	0.095^{***}	0.067^{*}
	(0.035)	(0.035)	(0.035)	(0.037)	(0.037)
Origin Precipitation (mm)		-0.014^{***}	-0.015^{***}	-0.034^{***}	-0.036^{***}
		(0.005)	(0.005)	(0.005)	(0.005)
Electricity Consumption (kWh)			0.110^{***}	0.074^{***}	0.092^{***}
			(0.013)	(0.012)	(0.013)
Refugee Share (%)				0.0003^{***}	0.0002^{**}
				(0.0001)	(0.0001)
Origin GDP pc ($)					-0.147^{***}
					(0.022)
Constant	5.440^{***}	5.513^{***}	4.630^{***}	4.908^{***}	5.979^{***}
	(0.179)	(0.178)	(0.205)	(0.222)	(0.279)
Observations	110,160	110,160	110,160	90,720	90,720
R-squared	0.741	0.741	0.742	0.743	0.746
Origin Fixed Effects	Yes	Yes	Yes	Yes	Yes
Destination Fixed Effects	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes
Control Variables	No	Yes	Yes	Yes	Yes

Note: The dependent variable is the logarithm of one plus internal migration flows between provinces. All explanatory variables enter the regressions in natural logarithmic form unless otherwise indicated. Temperature is measured in degrees Celsius (°C), precipitation in millimeters (mm), electricity consumption in kilowatt-hours (kWh) per capita, refugee share as the percentage of refugees in the origin province, and GDP pc in U.S. dollars ($). The squared temperature term captures potential nonlinearities in climate. The models follow a stepwise specification to evaluate the contribution of climate and socioeconomic factors to internal migration across the 81 provinces of Türkiye. Models (4)-(5) are estimated on the reduced sample for which refugee share data are available. All regressions include origin, destination, and year fixed effects. Robust standard errors clustered at the origin-destination pair level are reported in parentheses. Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$.

Model 4 introduces the refugee share variable. The sample size decreases because refugee data are available only for part of the study period. Refugee share enters with a positive and statistically significant coefficient, suggesting that provinces hosting larger refugee populations also experience higher rates of internal out-migration. Finally, Model 5 adds origin GDP per capita, which is negative and statistically significant, indicating lower out-migration from higher-income origin provinces. Although the magnitudes of the temperature coefficients decline as additional socioeconomic controls are introduced, both the linear and quadratic terms retain their expected signs and remain statistically significant throughout the sequential specifications. The persistent sign pattern indicates that the estimated nonlinear relationship between origin temperature and internal migration is robust to the inclusion of climatic and socioeconomic controls. As a robustness check, I also report the original specification based on mean-centered temperature measured in Kelvin, which avoids negative temperature values, in **??**, Table A.2

7.2.2 PPML Estimates: Full Sample

Table 3 reports the gravity model estimates obtained using PPML for the full sample of 81 provinces over the period 2008–2024. The specifications are estimated sequentially by progressively introducing climatic and socioeconomic controls. The dependent variable is migration flows between origin and destination provinces, estimated at the level using PPML. All specifications include origin, destination, and year fixed effects, and standard errors are clustered at the origin-destination pair level.

Model 1 includes the origin climate variables. The origin temperature coefficient is negative, whereas the quadratic temperature term is positive, and both coefficients are statistically significant. This sign pattern indicates a nonlinear relationship between origin temperature and migration flows, consistent with temperature extremes at the origin acting as push factors. Origin precipitation enters with a negative coefficient. Model 2 adds origin GDP per capita, which enters with a negative and statistically significant coefficient, while the estimated climate coefficients remain unchanged.

Table 3: PPML Estimates: Full Sample of 81 Provinces

	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Origin Temperature (°C)	-1.773***	-1.638***	-1.657***	-1.662***	-1.671***	-1.511***
	(0.305)	(0.306)	(0.303)	(0.300)	(0.298)	(0.326)
Origin Temperature^2	0.385***	0.363***	0.369***	0.371***	0.368***	0.309***
	(0.070)	(0.070)	(0.070)	(0.069)	(0.069)	(0.076)
Origin Precipitation (mm)	-0.021*	-0.019	-0.019*	-0.019*	-0.016	-0.025*
	(0.012)	(0.012)	(0.012)	(0.012)	(0.012)	(0.013)
Origin GDP pc ($)		-0.222***	-0.221***	-0.219***	-0.225***	-0.066
		(0.054)	(0.053)	(0.053)	(0.053)	(0.048)
Destination Temperature (°C)			0.024***	0.022***	0.021***	0.021***
			(0.006)	(0.006)	(0.006)	(0.007)
Destination Temperature^2			-0.001	-0.000	-0.001	-0.000
			(0.000)	(0.000)	(0.000)	(0.000)
Destination Precipitation (mm)			-0.019**	-0.021**	-0.019*	-0.031***
			(0.010)	(0.010)	(0.010)	(0.012)
Destination GDP pc ($)				0.164***	0.178***	0.189***
				(0.046)	(0.046)	(0.047)
Distance (km)					-0.764***	-0.768***
					(0.024)	(0.023)
Refugee Share (%)						0.000**
						(0.000)
Constant	9.125***	10.935***	11.043***	9.524***	14.278***	12.877***
	(0.331)	(0.556)	(0.549)	(0.683)	(0.659)	(0.726)
Observations	110,160	110,160	110,160	110,160	110,160	90,720
Pseudo R-squared	0.762	0.762	0.762	0.762	0.857	0.863
Origin Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Destination Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Control Variables	No	Yes	Yes	Yes	Yes	Yes

Note: The dependent variable is migration flows between origin and destination provinces and is estimated in levels using the PPML estimator. All explanatory variables enter the regressions in natural logarithmic form unless otherwise indicated. Temperature is measured in (°C), precipitation in (mm), GDP pc in ($), distance in (km), and refugee share in (%). The models follow a stepwise specification by progressively introducing climatic and socioeconomic controls. All specifications include fixed effects for origin, destination, and year. Robust standard errors clustered at the origin–destination pair level are reported in parentheses. Models (1)–(5) are estimated using the full sample, whereas Model (6) uses the reduced sample for which refugee share data are available. Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$.

Model 3 further introduces destination climate variables. The destination temperature term has a positive, statistically significant coefficient, whereas the quadratic term is not statistically significant, suggesting little evidence of a nonlinear effect of destination temperature. Destination precipitation enters with a negative coefficient. Model 4 adds destination GDP pc, which is positive and statistically significant, indicating that economically stronger destination provinces attract more migrants. Model 5 further includes geographical distance, which enters with a negative, statistically significant coefficient, consistent with the gravity model's prediction that migration declines with distance.

Finally, Model 6 introduces the refugee share variable. The sample size decreases because refugee data are available only for part of the study period. Refugee share enters with a positive and statistically significant coefficient. The inclusion of refugee share reduces both the magnitude and statistical significance of the origin GDP coefficient, indicating that the estimated income effect is sensitive to controlling for refugee presence. Overall, the estimated climate coefficients remain stable across specifications, indicating that the nonlinear relationship between origin temperature and migration is robust to the sequential inclusion of climatic, socioeconomic, and geographic controls. As a robustness check, the original specification based on mean-centered temperature measured in Kelvin is reported in Appendix A, Table A.3

7.2.3 Hausman-Taylor Estimates: Full Sample

Table 4 reports the Hausman-Taylor estimates for the full sample of 81 provinces over the period 2008–2024. The specifications are estimated sequentially by progressively introducing climatic and socioeconomic controls. All specifications include province-pair and year fixed effects, while the Hausman-Taylor estimator allows the estimation of the time-invariant distance variable. Model 1 includes origin temperature, its quadratic term, origin GDP pc, and geographical distance. The linear temperature coefficient is negative, whereas the quadratic temperature term is positive, and both coefficients are significant.

Table 4: Hausman–Taylor Estimates: Full Sample of 81 Provinces

	Model 1	Model 2	Model 3	Model 4	Model 5
Origin Temperature (°C)	-1.921***	-1.897***	-1.948***	-1.709***	-1.292***
	(0.102)	(0.102)	(0.146)	(0.145)	(0.144)
Origin Temperature^2	0.618***	0.614***	0.546***	0.456***	0.314***
	(0.021)	(0.021)	(0.030)	(0.030)	(0.030)
Origin Precipitation (mm)		-0.033***	0.024***	0.007	0.026***
		(0.004)	(0.005)	(0.005)	(0.005)
Wildfire Area (ha)			-0.006***	-0.005***	-0.004***
			(0.001)	(0.001)	(0.001)
Origin GDP pc ($)	0.124***	0.124***	0.075***	0.253***	0.184***
	(0.007)	(0.007)	(0.006)	(0.008)	(0.008)
Refugee Share (%)				0.220***	0.140***
				(0.007)	(0.007)
Electricity Consumption (kWh)					0.284***
					(0.007)
Distance (km)	-0.712***	-0.711***	-0.729***	-0.692***	-0.689***
	(0.025)	(0.025)	(0.024)	(0.024)	(0.024)
Constant	8.977***	9.152***	10.060***	7.829***	6.202***
	(0.212)	(0.214)	(0.251)	(0.257)	(0.261)
Observations	110,160	110,160	77,760	77,760	77,760
Province Pairs	6,480	6,480	6,480	6,480	6,480
Pair Fixed Effects	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes

Note: The dependent variable is the logarithm of internal migration flows between provinces. Estimation is performed using the Hausman–Taylor instrumental variables estimator, which allows the inclusion of time-invariant variables while addressing the endogeneity of selected regressors. All explanatory variables enter the regressions in natural logarithmic form unless otherwise indicated. Temperature is measured in (°C), precipitation in (mm), wildfire area in hectares (ha), electricity consumption in kilowatt-hours (kWh), GDP pc in ($), and distance in kilometers (km). Origin GDP per capita is treated as endogenous, whereas the remaining climate and socioeconomic variables are treated as exogenous. Geographic distance is identified through the internal instrumentation procedure of the Hausman-Taylor estimator. All specifications include province-pair and year fixed effects. Standard errors clustered at the province-pair level are reported in parentheses. Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$.

Model 2 adds origin precipitation, which enters with a negative and statistically significant coefficient. Model 3 further introduces wildfire area, which enters with a negative and statistically significant coefficient. Following the inclusion of the wildfire area, the estimated precipitation coefficient changes sign, suggesting that the estimated precipitation effect is sensitive to the inclusion of wildfire controls.

Model 4 adds the refugee share variable. The sample size decreases because wildfire and refugee data are available only for part of the study period. Refugee share enters with a positive and statistically significant coefficient. Origin GDP per capita is positive and statistically significant, whereas geographical distance has a negative, statistically significant coefficient, indicating that migration declines with distance.

Finally, Model 5 includes electricity consumption, which enters with a positive and statistically significant coefficient. Although the magnitude of the temperature coefficients declines as additional controls are introduced, both the linear and quadratic terms retain their expected signs and remain statistically significant across all specifications. This indicates that the estimated nonlinear relationship between origin temperature and migration is robust to the sequential inclusion of climatic and socioeconomic controls. As a robustness check, the original Hausman-Taylor specification, based on mean-centered temperature measured in Kelvin, is reported in Table A.4.

7.3 Climate-Exposed Provinces: 30–City Subsample

While the full-sample analysis provides a national overview of migration dynamics, this subsection focuses on a subsample of 30 climate-exposed provinces. Restricting the analysis to these provinces allows me to examine whether migration responses to climatic conditions are stronger in environmentally vulnerable regions. The empirical analysis follows the same approach as in the full-sample analysis. I first present the three-way fixed-effects estimates, followed by the PPML gravity model and the Hausman-Taylor estimator.

7.3.1 Three-Way Fixed Effects Estimates: Climate–Exposed Provinces

Table 5 reports the three-way fixed-effects estimates for the climate-exposed subsample of 30 provinces over the period 2008–2024. The models are estimated sequentially, with climatic and socioeconomic controls introduced progressively. All specifications include origin, destination, and year fixed effects, and standard errors are clustered at the origin–destination pair level.

Model 1 includes only the origin temperature and its quadratic term. Although the estimated coefficients retain the expected negative linear and positive quadratic signs, neither coefficient is statistically significant. Model 2 adds precipitation, which enters with a negative and statistically significant coefficient. Model 3 further includes electricity consumption, which is positive and statistically significant. Model 4 introduces the refugee share variable. The sample size decreases because refugee data are available only for part of the study period. Refugee share enters with a positive and statistically significant coefficient. Finally, Model 5 adds origin GDP pc, which is negative and statistically significant, while precipitation remains negative and statistically significant across all specifications. As a robustness check, the original specification based on mean-centered temperature measured in Kelvin is reported in **??**, Table A.5.

Table 5: Three-Way Fixed Effects Estimates: Climate-Exposed Subsample of 30 Provinces

	Model 1	Model 2	Model 3	Model 4	Model 5
Origin Temperature (°C)	-0.433	-0.068	0.220	0.350	0.156
	(0.704)	(0.711)	(0.708)	(0.806)	(0.808)
Origin Temperature^2	0.182	0.100	0.034	0.069	0.109
	(0.140)	(0.141)	(0.141)	(0.158)	(0.159)
Origin Precipitation (mm)		-0.056^{***}	-0.056^{***}	-0.076^{***}	-0.077^{***}
		(0.009)	(0.009)	(0.009)	(0.009)
Electricity Consumption (kWh)			0.064^{***}	0.026	0.046^{**}
			(0.017)	(0.017)	(0.018)
Refugee Share (%)				0.0004^{***}	0.0003^{***}
				(0.0001)	(0.0001)
Origin GDP pc ($)					-0.173^{***}
					(0.033)
Constant	4.534^{***}	4.509^{***}	3.734^{***}	3.584^{***}	5.195^{***}
	(0.890)	(0.887)	(0.896)	(1.023)	(1.048)
Observations	40,800	40,800	40,800	33,600	33,600
R-squared	0.729	0.729	0.729	0.730	0.730
Origin Fixed Effects	Yes	Yes	Yes	Yes	Yes
Destination Fixed Effects	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes
Control Variables	No	Yes	Yes	Yes	Yes

Note: The dependent variable is the logarithm of one plus internal migration flows between provinces. All explanatory variables enter the regressions in natural logarithmic form unless otherwise indicated. Temperature is measured in (°C), precipitation in (mm), electricity consumption in kilowatt-hours (kWh) per capita, refugee share as the percentage of refugees in the origin province, and GDP pc in ($). The squared temperature term captures potential nonlinearities in climate. The models follow a stepwise specification to evaluate the contribution of climate and socioeconomic factors to internal migration across the climate-exposed subsample of 30 provinces in Türkiye. Models (4)–(5) are estimated on the reduced sample for which refugee share data are available. All regressions include origin, destination, and year fixed effects. Robust standard errors clustered at the origin–destination pair level are reported in parentheses. Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$.

7.3.2 PPML Estimates: Climate-Exposed Provinces

Table 6 reports the gravity model estimates for the climate-exposed subsample of 30 origin provinces over the period 2008–2024. The specifications are estimated sequentially by progressively introducing climatic and socioeconomic controls. The dependent variable is migration flows between origin and destination provinces, estimated at the level using PPML. All specifications include origin, destination, and year fixed effects, and standard errors are clustered at the origin–destination pair level.

Model 1 includes the origin climate variables. The linear origin temperature coefficient is negative, whereas the quadratic temperature term is positive. The quadratic term is statistically significant at the 10 percent level, providing limited evidence of a nonlinear relationship between origin temperature and migration flows. Origin precipitation has a negative, statistically significant coefficient. Model 2 adds origin GDP pc, which enters with a negative and statistically significant coefficient, while the estimated origin climate coefficients remain largely unchanged.

Model 3 further introduces destination climate variables. The destination temperature term has a positive, statistically significant coefficient, whereas the quadratic destination temperature term is generally not statistically significant, suggesting little evidence of a nonlinear destination temperature effect. Destination precipitation enters with a negative coefficient. Model 4 adds destination GDP pc, which is positive and statistically significant.

Model 5 further includes geographical distance, which enters with a negative, statistically significant coefficient, consistent with the gravity model's prediction that migration declines with distance. Finally, Model 6 introduces the refugee share variable. The sample size decreases because refugee data are available only for part of the study period. Refugee share enters with a positive and statistically significant coefficient. The estimated origin temperature coefficients increase in magnitude and become statistically significant, providing stronger evidence of a nonlinear relationship between origin temperature and migration in the final specification.

Table 6: PPML Estimates: Climate-Exposed Subsample of 30 Provinces

	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Origin Temperature (°C)	-2.109	-2.047	-2.050	-2.049	-2.205	-3.857**
	(1.431)	(1.413)	(1.402)	(1.395)	(1.374)	(1.771)
Origin Temperature^2	0.485*	0.488*	0.490*	0.490*	0.513*	0.761**
	(0.287)	(0.282)	(0.280)	(0.279)	(0.274)	(0.341)
Origin Precipitation (mm)	-0.124***	-0.117***	-0.117***	-0.117***	-0.114***	-0.157***
	(0.026)	(0.026)	(0.025)	(0.025)	(0.025)	(0.031)
Origin GDP pc ($)		-0.265***	-0.265***	-0.261***	-0.269***	-0.071
		(0.079)	(0.078)	(0.078)	(0.078)	(0.063)
Destination Temperature (°C)			0.042***	0.039***	0.035***	0.035***
			(0.010)	(0.010)	(0.009)	(0.013)
Destination Temperature^2			-0.001*	-0.001	-0.001	-0.000
			(0.001)	(0.001)	(0.001)	(0.001)
Destination Precipitation (mm)			-0.011	-0.015	-0.002	-0.016
			(0.018)	(0.018)	(0.018)	(0.024)
Destination GDP pc ($)				0.269***	0.284***	0.224***
				(0.066)	(0.067)	(0.072)
Distance (km)					-1.141***	-1.129***
					(0.041)	(0.040)
Refugee Share (%)						0.001***
						(0.000)
Constant	9.576***	11.656***	11.674***	9.188***	16.586***	18.328***
	(1.724)	(1.725)	(1.732)	(1.894)	(1.885)	(2.468)
Observations	40,800	40,800	40,800	40,800	40,800	33,600
Pseudo R-squared	0.736	0.736	0.736	0.737	0.873	0.876
Origin Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Destination Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Control Variables	No	Yes	Yes	Yes	Yes	Yes

Note: The dependent variable is migration flows between origin and destination provinces and is estimated in levels using the PPML estimator. All explanatory variables enter the regressions in natural logarithmic form unless otherwise indicated. Temperature is measured in degrees (°C), precipitation in (mm), GDP per capita in ($), distance in (km), and refugee share as the percentage of refugees in the origin province. The models follow a stepwise specification by progressively introducing climatic and socioeconomic controls for the climate-exposed subsample of 30 origin provinces. All specifications include fixed effects for origin, destination, and year. Robust standard errors clustered at the origin–destination pair level are reported in parentheses. Models (1)–(5) are estimated using the full climate-exposed sample, whereas Model (6) uses the reduced sample for which refugee share data are available. Significance

For comparison, the corresponding PPML results for the climate-exposed subsample of 30 origin provinces using the original centered Kelvin specification are reported as a robustness check in Appendix A, Table A.6.

7.3.3 Hausman–Taylor Estimates: Climate-Exposed Provinces

Table 7 reports the Hausman-Taylor estimates for the climate-exposed subsample of 30 origin provinces over the period 2008–2024. The specifications are estimated sequentially by progressively introducing climatic and socioeconomic controls. All specifications include province-pair and year fixed effects, while the Hausman-Taylor estimator allows the estimation of the time-invariant distance variable.

Model 1 includes origin temperature, its quadratic term, origin GDP pc, and geographical distance. The linear temperature coefficient is negative, whereas the quadratic temperature term is positive, and both coefficients are statistically significant. This sign pattern indicates a nonlinear relationship between origin temperature and migration, consistent with temperature extremes at the origin acting as push factors. Origin GDP pc and geographical distance enter with negative, statistically significant coefficients, indicating that migration declines with distance. Model 2 adds origin precipitation, which enters with a negative and statistically significant coefficient.

Model 3 further introduces wildfire area, which enters with a positive and statistically significant coefficient. The estimated precipitation coefficient remains negative and statistically significant, while the temperature coefficients continue to retain their expected signs and statistical significance. The sample size decreases because wildfire data are available only for part of the study period. Model 4 adds the refugee share variable, which is not statistically significant.

Table 7: Hausman–Taylor Estimates: Climate-Exposed Subsample of 30 Provinces

	Model 1	Model 2	Model 3	Model 4	Model 5
Origin Temperature (°C)	-1.044***	-0.831**	-2.025***	-1.989***	-1.908***
	(0.356)	(0.358)	(0.540)	(0.555)	(0.558)
Origin Temperature^2	0.350***	0.303***	0.568***	0.560***	0.541***
	(0.072)	(0.073)	(0.106)	(0.110)	(0.111)
Origin Precipitation (mm)		-0.042***	-0.087***	-0.088***	-0.089***
		(0.008)	(0.009)	(0.009)	(0.009)
Wildfire Area (ha)			0.008***	0.009***	0.009***
			(0.002)	(0.002)	(0.002)
Origin GDP pc ($)	-0.244***	-0.237***	-0.077***	-0.074***	-0.108***
	(0.021)	(0.021)	(0.024)	(0.024)	(0.025)
Refugee Share (%)				0.004	-0.0004
				(0.014)	(0.014)
Electricity Consumption (kWh)					0.026*
					(0.014)
Distance (km)	-0.601***	-0.598***	-0.616***	-0.615***	-0.614***
	(0.040)	(0.041)	(0.039)	(0.039)	(0.039)
Constant	10.670***	10.610***	11.300***	11.220***	11.290***
	(0.540)	(0.540)	(0.776)	(0.807)	(0.810)
Observations	40,800	40,800	28,800	28,800	28,800
Province Pairs	2,400	2,400	2,400	2,400	2,400
Pair Fixed Effects	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes

Note: The dependent variable is the logarithm of internal migration flows between provinces. Estimation is performed using the Hausman-Taylor instrumental variables estimator, which allows the inclusion of time-invariant variables while addressing the endogeneity of selected regressors. All explanatory variables enter the regressions in natural logarithmic form unless otherwise indicated. Temperature is measured in (°C), precipitation in (mm), wildfire area in hectares (ha), electricity consumption in kilowatt-hours (kWh), GDP pc in ($), and distance in (km). The sample is restricted to the 30 origin provinces identified as highly exposed to climatic stress. Origin GDP pc is treated as endogenous, whereas the remaining climate and socioeconomic variables are treated as exogenous. Geographic distance is identified through the internal instrumentation procedure of the Hausman-Taylor estimator. All specifications include province-pair and year fixed effects. Standard errors clustered at the province-pair level are reported in parentheses. Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$.

Finally, Model 5 includes electricity consumption, which enters with a positive coefficient and is statistically significant at the 10 percent level. Although the magnitude of the temperature coefficients varies across specifications, both the linear and quadratic terms remain statistically significant throughout, indicating a robust nonlinear relationship between origin temperature and migration within the climate-exposed subsample. As a robustness check, the original specification based on mean-centered temperature measured in Kelvin is reported in Appendix A, Table A.7.

7.4 Drought-Exposed Provinces: 17-City Subsample

To further examine the relationship between climate conditions and internal migration, this subsection focuses on a subsample of 17 provinces identified as highly exposed to drought conditions. Restricting the analysis to these provinces allows me to examine whether migration responses are stronger in regions facing the most severe climatic pressures.

The empirical analysis follows the same approach as in the previous sections. I first present the three-way fixed-effects estimates, followed by the PPML gravity model and the Hausman-Taylor estimator.

7.4.1 Three-Way Fixed Effects Estimates: Drought–Exposed Provinces

Table 8 reports the three-way fixed-effects estimates for the drought-exposed subsample of 17 origin provinces over the period 2008–2024. The specifications are estimated sequentially by progressively introducing climatic and socioeconomic controls. All specifications include origin, destination, and year fixed effects, and standard errors are clustered at the origin–destination pair level.

Model 1 includes only the origin temperature and its quadratic term. The linear temperature coefficient is negative, whereas the quadratic term is positive, and both coefficients are statistically significant. This sign pattern indicates a nonlinear relationship between origin temperature and migration, consistent with temperature extremes acting as push factors.

Table 8: Three-Way Fixed Effects Estimates: Drought-Exposed Subsample of 17 Provinces

	Model 1	Model 2	Model 3	Model 4	Model 5
Origin Temperature (°C)	-3.799^{***}	-3.174^{***}	-2.867^{**}	-3.173^{**}	-2.436^{*}
	(1.121)	(1.137)	(1.124)	(1.343)	(1.353)
Origin Temperature^2	0.737^{***}	0.616^{***}	0.549^{***}	0.623^{**}	0.502^{**}
	(0.212)	(0.215)	(0.213)	(0.252)	(0.253)
Origin Precipitation		-0.071^{***}	-0.075^{***}	-0.085^{***}	-0.079^{***}
		(0.012)	(0.012)	(0.012)	(0.012)
Electricity Consumption (kWh)			0.071^{***}	0.040^{*}	0.054^{**}
			(0.023)	(0.022)	(0.023)
Refugee Share (%)				0.0004^{***}	0.0003^{***}
				(0.0001)	(0.0001)
Origin GDP pc ($)					-0.155^{***}
					(0.041)
Constant	9.644^{***}	9.300^{***}	8.467^{***}	9.073^{***}	9.176^{***}
	(1.486)	(1.492)	(1.474)	(1.788)	(1.801)
Observations	23,120	23,120	23,120	19,040	19,040
R-squared	0.825	0.825	0.826	0.825	0.826
Origin Fixed Effects	Yes	Yes	Yes	Yes	Yes
Destination Fixed Effects	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes
Control Variables	No	Yes	Yes	Yes	Yes

Note: The dependent variable is the logarithm of one plus internal migration flows between provinces. All explanatory variables enter the regressions in natural logarithmic form unless otherwise indicated. Temperature is measured in (°C), precipitation in (mm), electricity consumption pc in kilowatt-hours (kWh), refugee share as a percentage of refugees in the origin province, and GDP pc in ($). The squared temperature term captures potential nonlinearities in climate. The models follow a stepwise specification to evaluate the contribution of climate and socioeconomic factors to internal migration across the drought-exposed subsample of 17 provinces in Türkiye. Models (4)–(5) are estimated on the reduced sample for which refugee share data are available. Wildfire is excluded from the preferred specification because its inclusion substantially reduces the sample size and produces unstable inference. All regressions include origin, destination, and year fixed effects. Robust standard errors clustered at the origin–destination pair level are reported in parentheses. Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$.

Model 2 adds precipitation, which enters with a negative and statistically significant coefficient, while the estimated temperature coefficients remain statistically significant. Model 3 further includes electricity consumption, which is positive and statistically significant, whereas precipitation remains negative and statistically significant. Model 4 introduces the refugee share variable. The sample size decreases because refugee data are available only for part of the study period. Refugee share enters with a positive and statistically significant coefficient, while electricity consumption remains positive and statistically significant.

Finally, Model 5 adds origin GDP per capita, which enters with a negative and statistically significant coefficient. Although the magnitude of the temperature coefficients declines as additional controls are introduced, both the linear and quadratic terms remain statistically significant across all specifications, indicating a robust nonlinear relationship between origin temperature and migration. As a robustness check, the original specification based on mean-centered temperature measured in Kelvin is reported in Appendix A, Table A.8.

7.4.2 PPML Estimates: Drought-Exposed Provinces

Table 9 reports the gravity model estimates for the drought-exposed subsample of 17 origin provinces over the period 2008–2024. The specifications are estimated sequentially by progressively introducing climatic and socioeconomic controls. The dependent variable is migration flows between origin and destination provinces, estimated at the level using PPML. All specifications include origin, destination, and year fixed effects, and standard errors are clustered at the origin–destination pair level.

Model 1 includes the origin climate variables. The linear origin temperature coefficient is negative and statistically significant, whereas the quadratic temperature term is positive and statistically significant. This sign pattern indicates a nonlinear relationship between origin temperature and migration flows, consistent with temperature extremes at the origin acting as push factors. Origin precipitation has a negative, significant coefficient.

Table 9: PPML Estimates: Drought-Exposed Subsample of 17 Provinces

	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Origin Temperature (°C)	-0.549**	-0.333	-0.327	-0.330	-0.344	-1.212***
	(0.233)	(0.254)	(0.254)	(0.253)	(0.252)	(0.315)
Origin Temperature^2	0.166***	0.161***	0.161***	0.161***	0.157***	0.254***
	(0.048)	(0.048)	(0.048)	(0.048)	(0.048)	(0.059)
Origin Precipitation (mm)	-0.143***	-0.123***	-0.122***	-0.122***	-0.123***	-0.171***
	(0.030)	(0.028)	(0.028)	(0.028)	(0.028)	(0.030)
Origin GDP pc ($)		-0.285***	-0.285***	-0.280***	-0.284***	0.024
		(0.098)	(0.097)	(0.097)	(0.097)	(0.082)
Destination Temperature (°C)			0.032***	0.030**	0.027**	0.026**
			(0.012)	(0.012)	(0.012)	(0.013)
Destination Temperature^2			-0.002*	-0.002*	-0.002*	-0.001
			(0.001)	(0.001)	(0.001)	(0.001)
Destination Precipitation (mm)			0.014	0.012	0.017	0.021
			(0.023)	(0.023)	(0.023)	(0.028)
Destination GDP pc ($)				0.201**	0.204**	0.144
				(0.083)	(0.084)	(0.088)
Distance (km)					-0.987***	-0.987***
					(0.065)	(0.064)
Refugee Share (%)						0.001***
						(0.000)
Constant	7.139***	8.952***	8.808***	6.960***	13.278***	12.641***
	(0.401)	(0.755)	(0.745)	(0.970)	(1.005)	(1.282)
Observations	23,120	23,120	23,120	23,120	23,120	19,040
Pseudo R-squared	0.752	0.762	0.782	0.796	0.854	0.896
Origin Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Destination Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Control Variables	No	Yes	Yes	Yes	Yes	Yes

Note: The dependent variable is migration flows between origin and destination provinces and is estimated in levels using PPML. All explanatory variables enter the regressions in natural logarithmic form unless otherwise indicated. Temperature is measured in (°C), precipitation in (mm), GDP pc in ($), distance in (km), and refugee share as the percentage of refugees in the origin province. The models follow a stepwise specification by progressively introducing climatic and socioeconomic controls for the drought-exposed subsample of 17 origin provinces. All specifications include fixed effects for origin, destination, and year. Robust standard errors clustered at the origin–destination pair level are reported in parentheses. Models (1)–(5) are estimated using the full drought-exposed sample, whereas Model (6)

Model 2 adds origin GDP per capita, which enters with a negative and statistically significant coefficient. Although the linear temperature coefficient is no longer statistically significant, the quadratic temperature term remains positive and statistically significant. Model 3 further introduces destination climate variables. Destination temperature enters with a positive and statistically significant coefficient, whereas the quadratic destination temperature term is statistically significant in Models 3–5. Destination precipitation enters with a positive but statistically insignificant coefficient.

Model 4 adds destination GDP per capita, which is positive and statistically significant. Model 5 further includes geographical distance, which enters with a negative and statistically significant coefficient. Finally, Model 6 introduces the refugee share variable. The sample size decreases because refugee data are available only for part of the study period. Refugee share enters with a positive and statistically significant coefficient. In the final specification, the linear origin temperature coefficient becomes statistically significant again, while the positive quadratic term remains statistically significant throughout the analysis. As a robustness check, the original specification based on mean-centered temperature measured in Kelvin is reported in Appendix A, Table A.9.

7.4.3 Hausman–Taylor Estimates: Drought-Exposed Provinces

Table 10 reports the Hausman–Taylor estimates for the drought-exposed subsample of 17 origin provinces over the period 2008–2024. The specifications are estimated sequentially by progressively introducing climatic and socioeconomic controls. All specifications include province–pair and year fixed effects, while the Hausman–Taylor estimator allows the estimation of the time-invariant distance variable.

Model 1 includes origin temperature, its quadratic term, origin GDP pc, and geographical distance. The linear temperature coefficient is negative, whereas the quadratic temperature term is positive, and both coefficients are statistically significant. This sign pattern indicates a nonlinear relationship between origin temperature and migration, consistent with temperature extremes at the origin acting as push factors. Origin GDP per capita and geographical distance enter with negative, statistically significant coefficients, indicating that migration declines with distance. Model 2 adds origin precipitation, which enters with a negative and statistically significant coefficient.

Model 3 further introduces wildfire area, which enters with a positive and statistically significant coefficient, while the estimated temperature coefficients remain statistically significant. The sample size decreases because wildfire data are available only for part of the study period. Model 4 adds the refugee share variable, which enters with a positive and statistically significant coefficient. Finally, Model 5 includes electricity consumption, which enters with a positive and statistically significant coefficient. Throughout all specifications, the temperature coefficients remain large and statistically significant, providing strong evidence of a nonlinear relationship between origin temperature and migration within the drought-exposed provinces. As a robustness check, the original specification based on mean-centered temperature, measured in Kelvin, is reported in Appendix A, Table A.10.

Overall, the empirical results indicate that climate conditions play an important role in shaping internal migration in Türkiye, although the estimated effects vary across samples and estimation methods. The full-sample analysis provides evidence of a nonlinear relationship between origin temperature and migration, while the subsample analyses reveal substantial regional heterogeneity. In particular, the temperature effects are strongest and most robust in the drought-exposed provinces, suggesting that migration responses intensify under more severe climatic conditions. Across the different specifications, precipitation, geographical distance, refugee presence, and socioeconomic characteristics also contribute to explaining migration patterns. As an additional robustness check, Table A.11 reports a temperature-threshold specification based on discrete temperature categories.

Table 10: Hausman-Taylor Estimates: Drought-Exposed Subsample of 17 Provinces

	Model 1	Model 2	Model 3	Model 4	Model 5
Origin Temperature (°C)	-3.899***	-3.607***	-6.028***	-4.878***	-4.336***
	(0.692)	(0.696)	(1.278)	(1.278)	(1.286)
Origin Temperature^2	0.829***	0.772***	1.260***	1.012***	0.899***
	(0.135)	(0.135)	(0.236)	(0.237)	(0.239)
Origin Precipitation (mm)		-0.044***	-0.071***	-0.077***	-0.080***
		(0.011)	(0.012)	(0.012)	(0.012)
Wildfire Area (ha)			0.009***	0.008***	0.008***
			(0.002)	(0.002)	(0.002)
Origin GDP pc ($)	-0.288***	-0.274***	-0.176***	-0.146***	-0.172***
	(0.026)	(0.026)	(0.031)	(0.032)	(0.032)
Refugee Share (%)				0.117***	0.115***
				(0.021)	(0.021)
Electricity Consumption (kWh)					0.054***
					(0.017)
Distance (km)	-0.660***	-0.657***	-0.679***	-0.649***	-0.643***
	(0.055)	(0.055)	(0.054)	(0.060)	(0.061)
Constant	15.630***	15.400***	18.320***	16.310***	15.420***
	(0.944)	(0.946)	(1.757)	(1.783)	(1.801)
Observations	23,120	23,120	16,320	16,320	16,320
Province Pairs	1,360	1,360	1,360	1,360	1,360
Pair Fixed Effects	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes

Note: The dependent variable is the logarithm of internal migration flows between provinces. Estimation is performed using the Hausman-Taylor instrumental variables estimator, which allows the inclusion of time-invariant variables while controlling for endogeneity in the selected regressors. All explanatory variables enter the regressions in natural logarithmic form unless otherwise indicated. Temperature is measured in (°C), precipitation in (mm), wildfire area in (ha), electricity consumption in (kWh), GDP pc in ($), and distance in (km). The sample is restricted to the 17 origin provinces identified as highly exposed to drought conditions. Origin GDP pc is treated as endogenous, whereas the remaining climate and socioeconomic variables are treated as exogenous. Geographic distance is identified through the internal instrumentation procedure of the Hausman-Taylor estimator. All specifications include province-pair and year fixed effects. Standard errors clustered at the province–pair level are reported in parentheses. Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$.

8 Discussion

In this chapter, I examine internal migration across Türkiye as a function of climatic conditions, economic opportunities, and demographic pressures within a gravity framework. Across the three estimation approaches—three-way fixed effects, PPML, and Hausman–Taylor—I find consistent evidence that climatic stress, together with economic and demographic factors, plays an important role in shaping internal migration patterns. Among the climatic variables considered, temperature emerges as the most influential determinant of migration.

The central finding is the nonlinear effect of temperature on migration. The estimated quadratic temperature term remains positive and statistically significant across most specifications, particularly in the full sample and the drought-exposed subsamples. This pattern suggests that migration responses become stronger as temperatures move away from moderate climatic conditions. While moderate increases in temperature appear to have relatively limited effects on migration decisions, extreme heat substantially increases migration pressures. The consistency of this pattern across the three estimation approaches strengthens confidence that the result is not driven by a particular econometric specification. These findings are consistent with previous evidence documenting nonlinear migration responses to temperature (Bohra-Mishra, Oppenheimer, and Hsiang, 2014; Carleton, Hsiang, and Burke, 2016; Missirian and Schlenker, 2017). In the Turkish context, where several southern provinces already experience high summer temperatures, additional warming appears to act as an important push factor, encouraging migration toward more temperate destinations (Backhaus, Martinez-Zarzoso, and Muris, 2015; Cai et al., 2016).

The temperature-bin specification provides additional evidence on the nonlinear relationship between temperature and migration. The estimated results indicate that migration rates increase sharply once summer maximum temperatures exceed approximately 32°C. Below this threshold, migration responses remain relatively modest, suggesting that households can adapt to ordinary climatic fluctuations. Beyond it, however, migration becomes substantially more likely. Consistent with these findings, the temperature-bin specification also indicates that extreme summer temperatures in origin provinces act as strong push factors, while relatively mild winter conditions increase the attractiveness of destination provinces. Taken together, the quadratic and temperature-bin specifications suggest that migration responses become substantially stronger once temperatures reach sufficiently high levels, highlighting the importance of accounting for nonlinear climate effects when analyzing internal migration.

I also find that climatic conditions interact with broader economic and demographic forces. Economic opportunities remain important determinants of migration, as provinces with stronger economic performance continue to attract migrants. At the same time, climatic conditions influence migration independently of these economic factors, indicating that environmental stress affects migration decisions through channels that extend beyond labor-market opportunities alone. The stronger climate effects observed in the drought-exposed provinces further suggest that environmental vulnerability amplifies migration responses to climatic shocks.

Demographic pressures also play an important role. I find evidence that refugee inflows are associated with changes in internal migration patterns, although this relationship is less robust across specifications than the nonlinear temperature effect. Nevertheless, the results suggest that population movements driven by conflict and displacement may indirectly influence the mobility decisions of local residents. Given Türkiye's position as one of the world's largest refugee-hosting countries, this finding highlights the importance of considering demographic and environmental pressures jointly rather than in isolation.

Several limitations should also be acknowledged. First, although the three-way fixed-effects, PPML, and Hausman–Taylor specifications control for a broad set of observable and unobservable factors, the estimated relationships should be interpreted as conditional associations rather than fully causal effects. Second, the climate-exposed and drought-exposed subsamples are constructed to examine heterogeneous responses to environmental stress and may not be fully representative of all provinces in Türkiye. Finally, the refugee channel remains indirect. Climate change may contribute to displacement through a combination of economic, environmental, and political mechanisms, making it difficult to isolate the precise pathways linking climate conditions, refugee movements, and internal migration.

Taken together, these findings indicate that climate variability, economic conditions, demographic pressures, and migration dynamics are closely interconnected. Rising temperatures appear to increase migration pressures, particularly in environmentally vulnerable regions, while refugee inflows and economic conditions further shape mobility patterns. These results underscore the importance of integrated policy approaches that simultaneously address environmental risks, economic adaptation, and demographic pressures when designing strategies to manage internal migration in the context of climate change.

9 Conclusion

In this chapter, I examine the relationship between climatic conditions and internal migration in Türkiye using a gravity framework grounded in the Random Utility Model (RUM) and province-level migration data covering the period 2008–2024. The analysis provides evidence that climatic conditions, alongside economic and demographic factors, play an important role in shaping migration decisions. The main findings are broadly consistent across the three-way fixed-effects, PPML, and Hausman–Taylor estimators, increasing confidence that they are not driven by a particular estimation approach.

The clearest finding is the nonlinear effect of temperature on migration. Migration responses become stronger as temperatures move away from moderate levels, while the temperature-bin specification suggests a threshold of approximately 32°C, beyond which migration rates increase sharply. These findings indicate that extreme heat acts as a powerful push factor, particularly in provinces already exposed to climatic stress. Precipitation also appears to influence migration, especially in drought-prone regions where agricultural activity remains an important source of income and employment.

I also find evidence that demographic pressures associated with refugee inflows influence internal migration patterns. Although refugee movements are not the primary focus of this study, the results suggest that higher refugee shares are associated with larger migration flows in several specifications, indicating that environmental and demographic pressures may interact in shaping mobility decisions. In the Turkish context, the large inflow of Syrian refugees has altered regional population dynamics and may have contributed to migration pressures in provinces already facing climatic challenges.

The climate-exposed and drought-exposed subsamples reinforce these conclusions. Migration responses to both temperature and precipitation are generally stronger in environmentally vulnerable provinces, highlighting the importance of regional exposure and adaptive capacity in shaping migration outcomes.

Taken together, the findings indicate that climate variability, demographic pressures, and economic conditions jointly influence internal migration in Türkiye. As temperatures continue to rise and drought conditions intensify in parts of the country, migration pressures may become increasingly concentrated in environmentally vulnerable regions. Future research could extend this analysis using higher-frequency climate data, household-level migration data, and explicit measures of adaptation to better understand how individuals respond to environmental change.

Appendix A Additional Figures and Tables

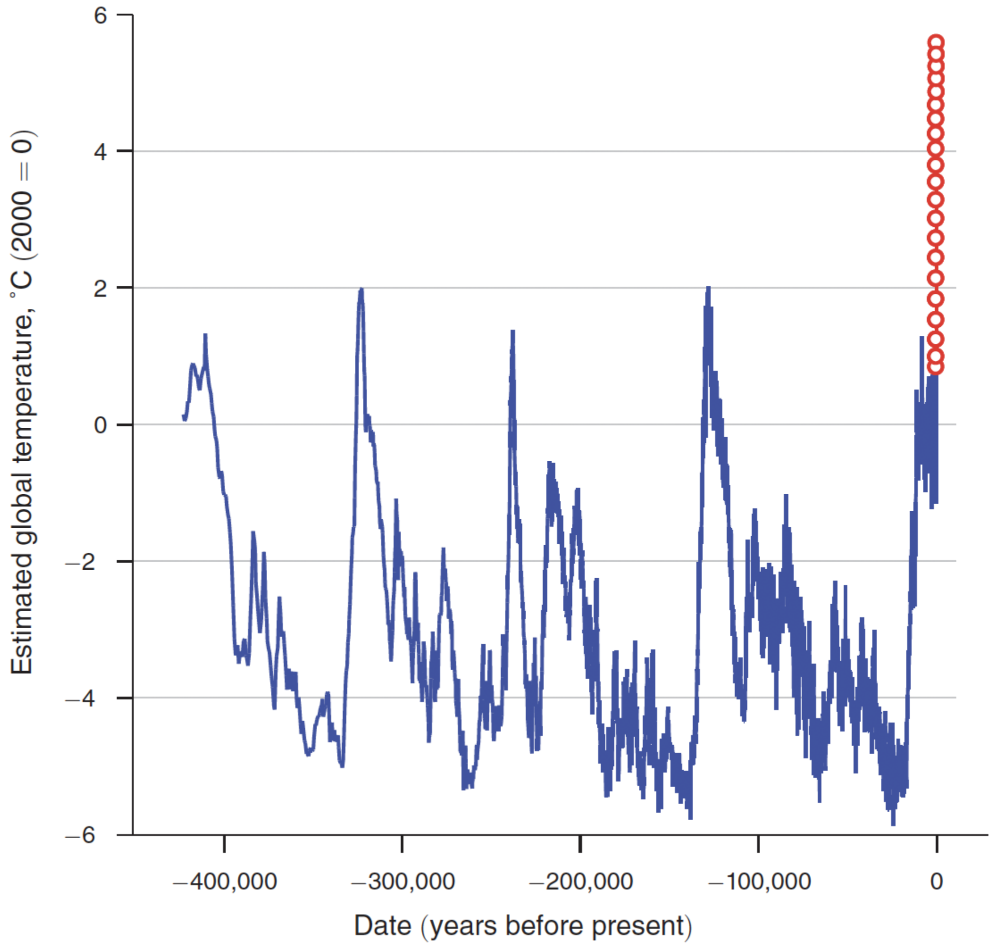

Figure A.1: Long-Run Global Temperature Variability and Future Projections

Note: The figure reports reconstructed global temperature changes over approximately 400,000 years using Antarctic ice-core data (solid line) together with projected temperature changes for the next two centuries (circles). Temperatures are normalized relative to the year 2000. Data source: Nordhaus (2013) and Nordhaus (2019).

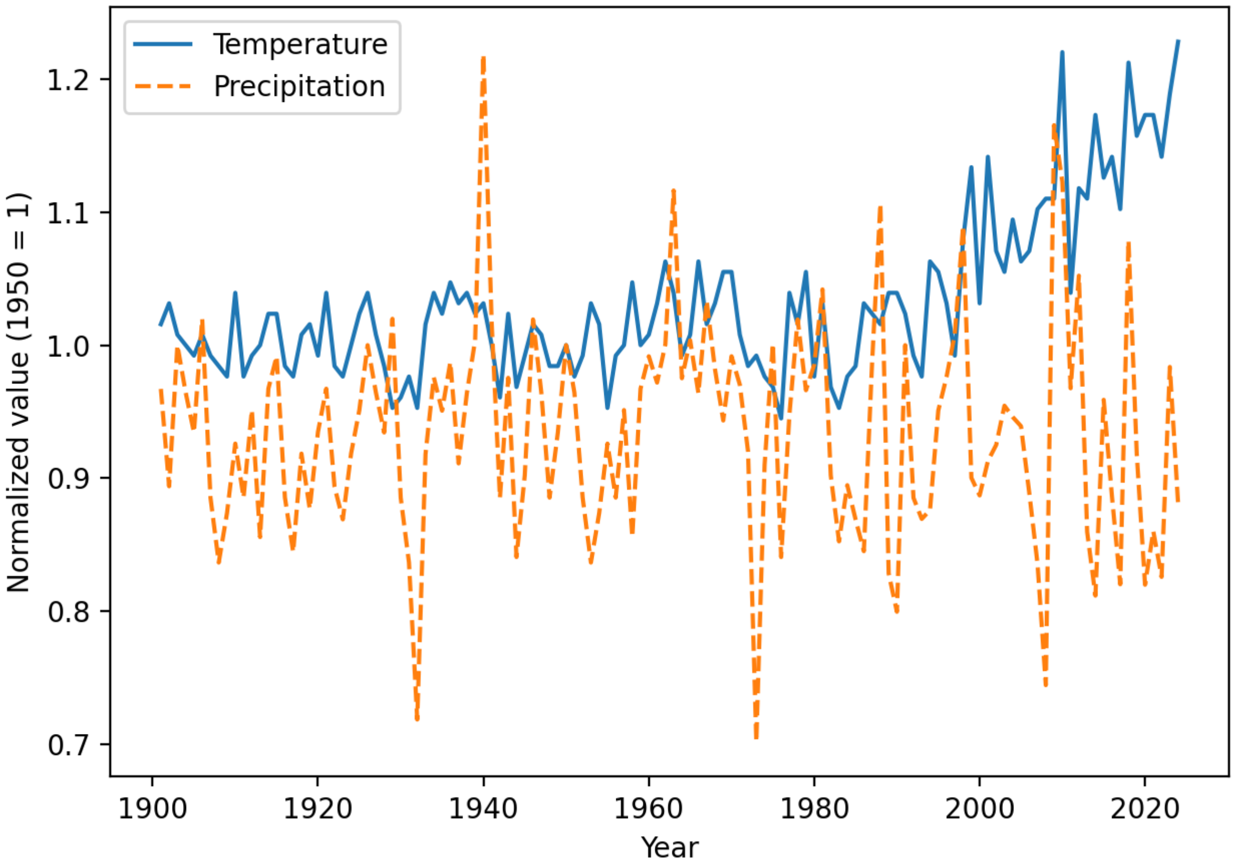

Figure A.2: Long-run trends in average temperature and precipitation in Türkiye over the period 1901–2024.

*Note:*Both series are normalized to their 1950 values to facilitate comparison across variables measured in different units.

Figure A.3: Trends in average temperature and precipitation in Türkiye for the period 1970–2024, normalized to 1950.

Note: The figure highlights the acceleration of temperature increases in the post-1970 period.

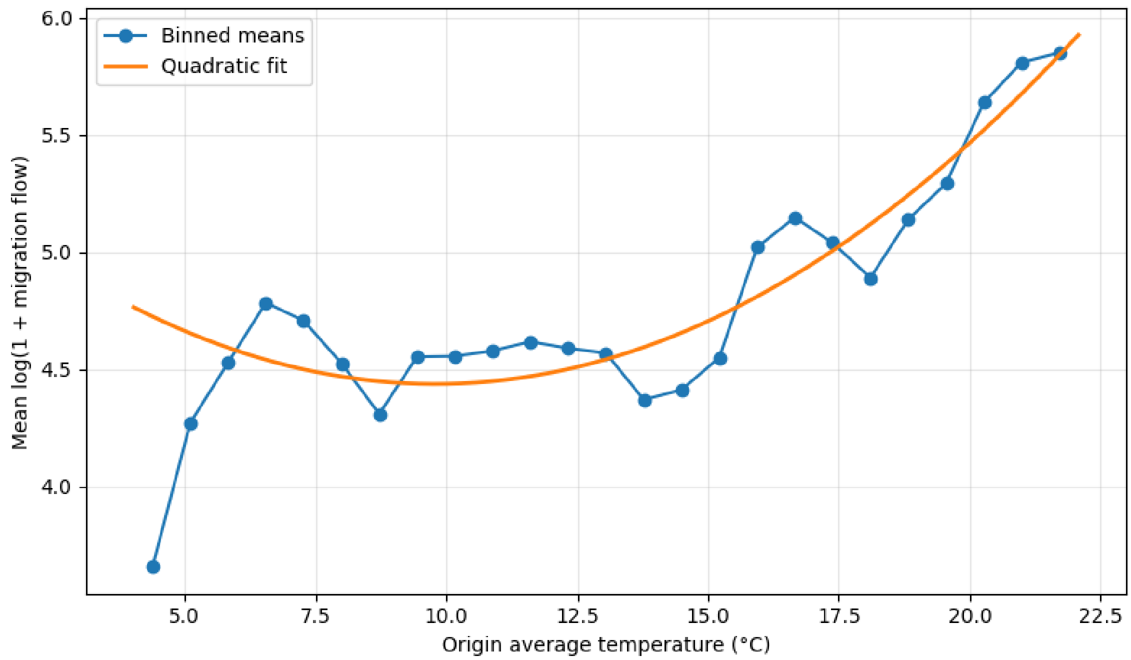

Figure A.4: Migration Flow and Origin Average Temperature

Note: This figure plots binned averages of the logarithm of one plus migration flows against origin average temperature. The solid line represents a quadratic fit.

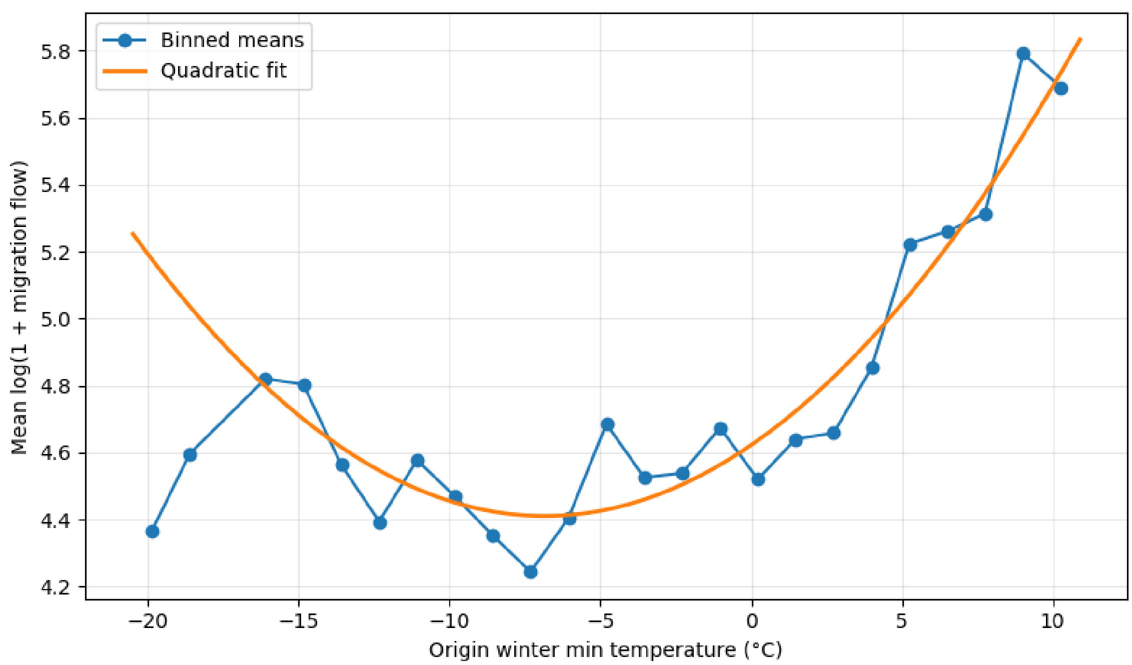

Figure A.5: Migration Flow and Origin Winter Minimum Temperature

Note: This figure plots binned averages of the logarithm of one plus migration flows against origin winter minimum temperature. The solid line represents a quadratic fit.

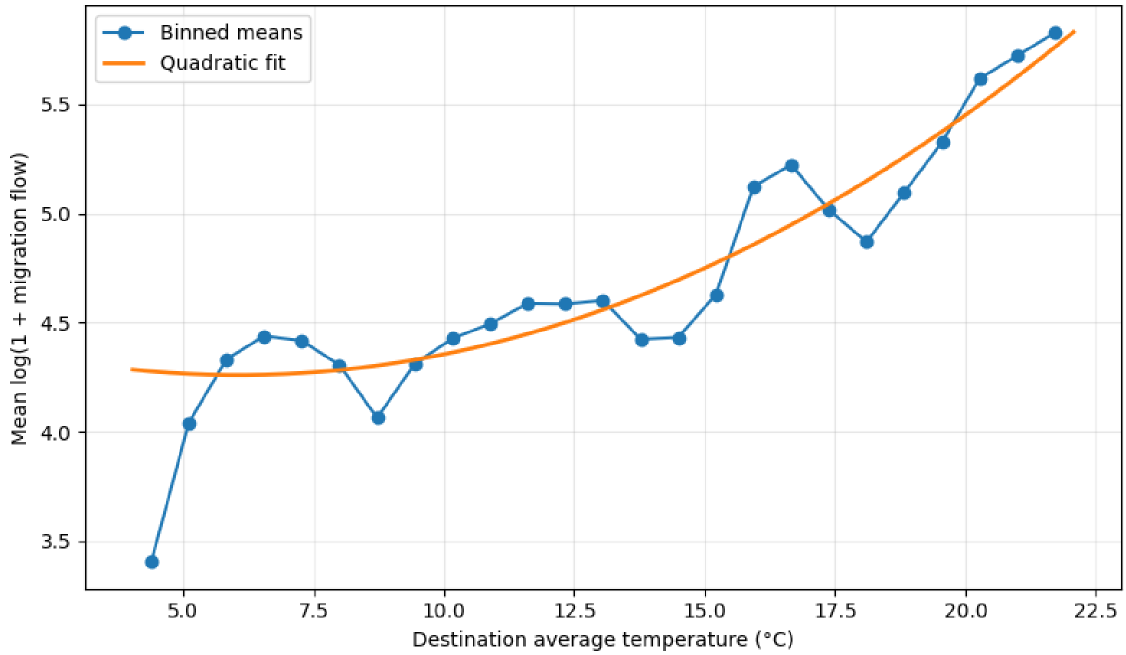

Figure A.6: Migration Flow and Destination Average Temperature

Note: This figure plots binned averages of the logarithm of one plus migration flows against destination average temperature. The solid line represents a quadratic fit.

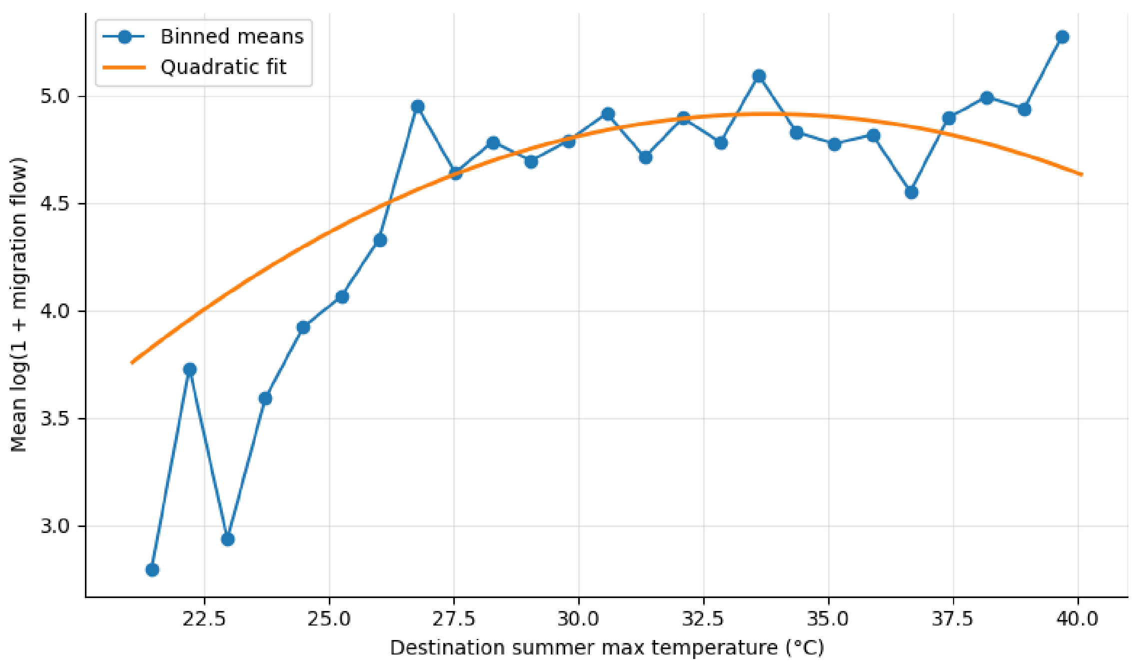

Figure A.7: Migration Flow and Destination Summer Maximum Temperature

Note: This figure plots binned averages of the logarithm of one plus migration flows against destination summer maximum temperature. The solid line represents a quadratic fit.

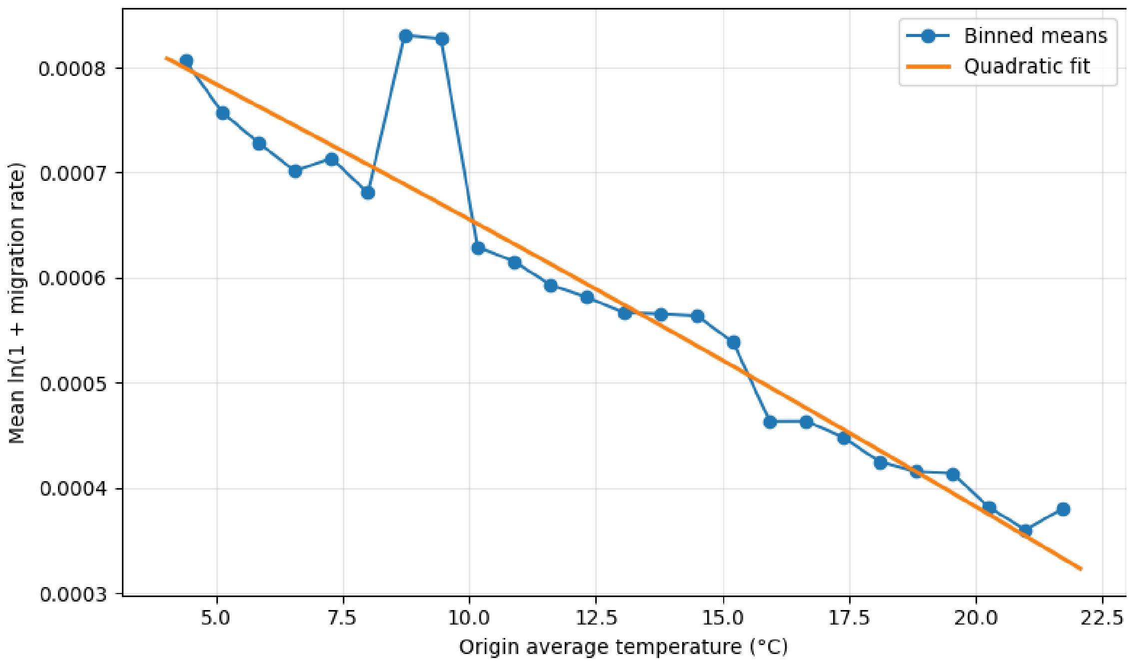

Figure A.8: Migration Rate and Origin Average Temperature

Note: This figure plots binned averages of the logarithm of one plus the migration rate against origin average temperature. Migration rates are defined as migration flows per capita in the origin country. The solid line represents a quadratic fit.

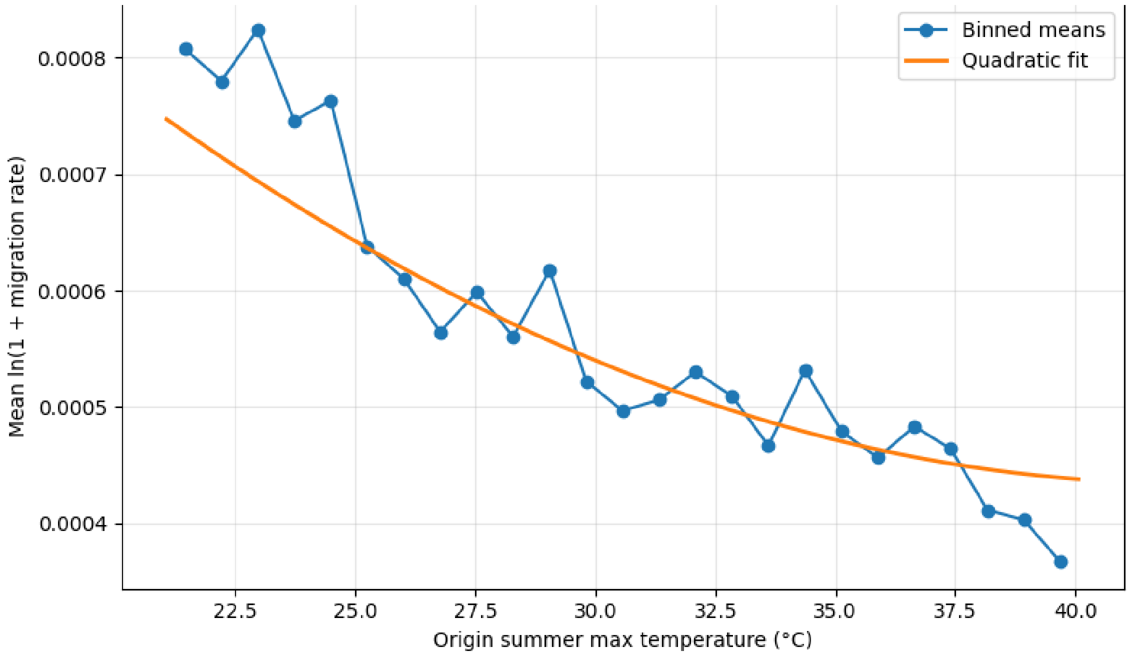

Figure A.9: Migration Rate and Origin Summer Maximum Temperature

Note: This figure plots binned averages of the logarithm of one plus the migration rate against origin summer maximum temperature. Migration rates are defined as migration flows divided by origin population. The solid line represents a quadratic fit.

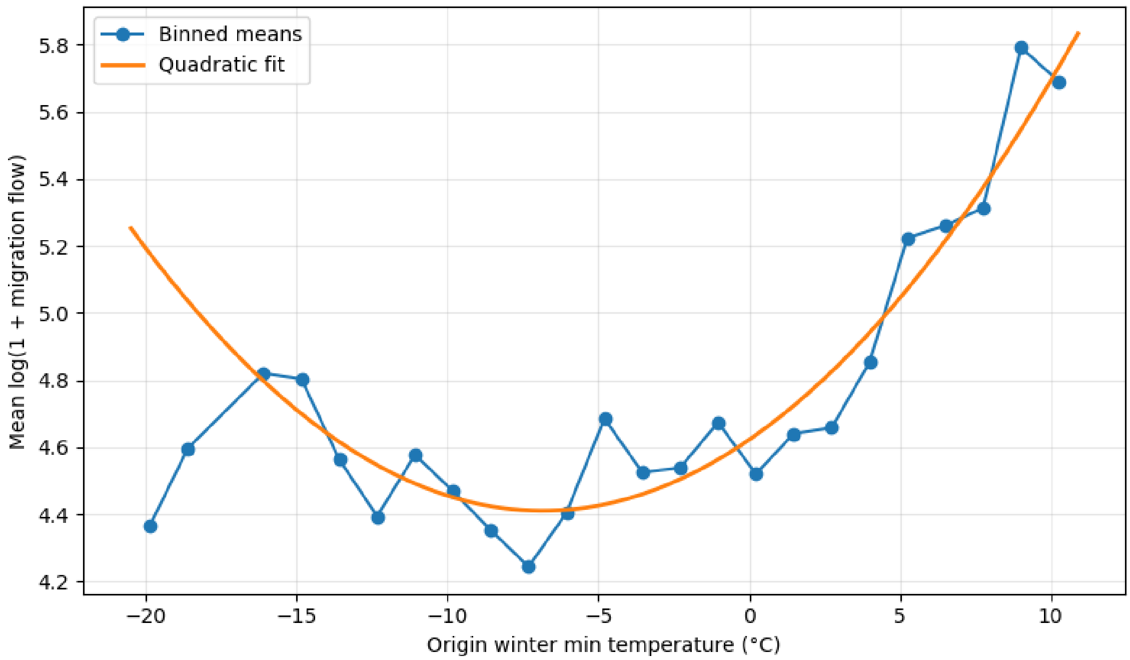

Figure A.10: Migration Rate and Origin Winter Minimum Temperature

Note: This figure plots binned averages of the logarithm of one plus the migration rate against origin winter minimum temperature. Migration rates are defined as migration flows divided by origin population. The solid line represents a quadratic fit.

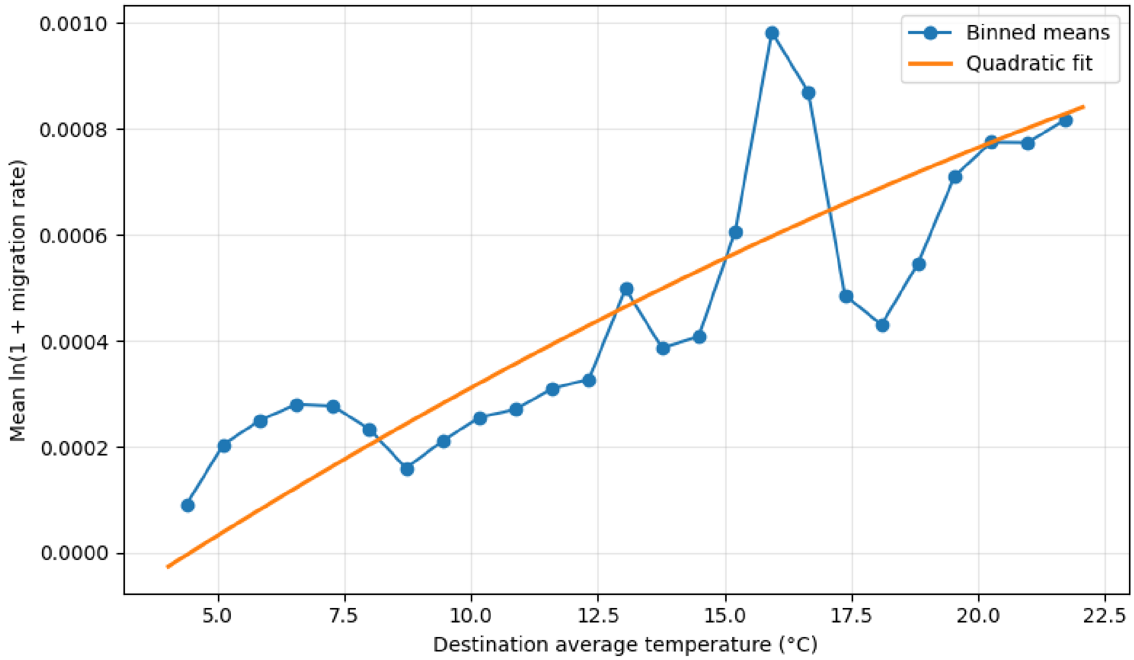

Figure A.11: Migration Rate and Destination Average Temperature

Note: This figure plots binned averages of the logarithm of one plus the migration rate against destination average temperature. Migration rates are defined as migration flows divided by origin population. The solid line represents a quadratic fit.

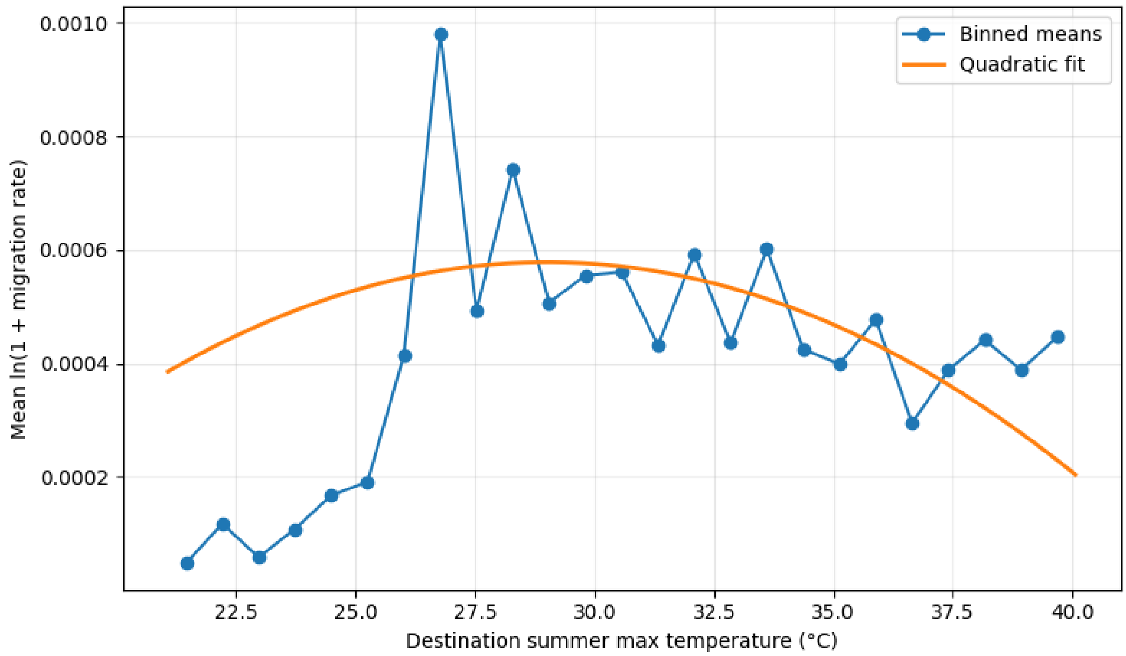

Figure A.12: Migration Rate and Destination Summer Maximum Temperature

Note: This figure plots binned averages of the logarithm of one plus the migration rate against destination summer maximum temperature. Migration rates are defined as migration flows divided by the population of the origin country. The solid line represents a quadratic fit.

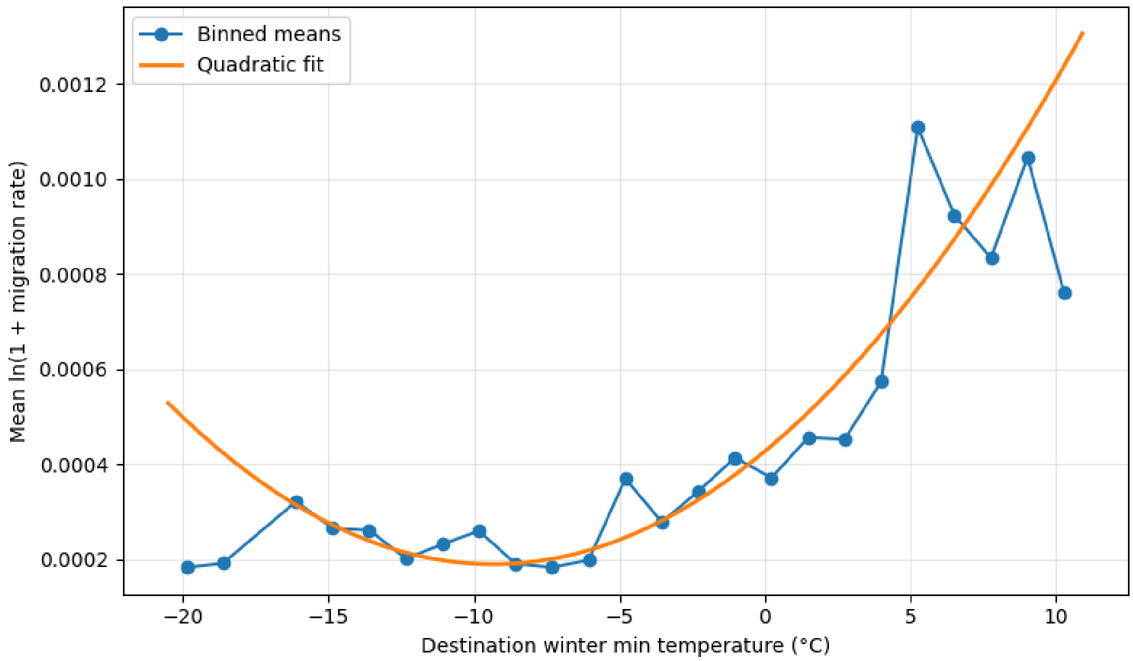

Figure A.13: Migration Rate and Destination Winter Minimum Temperature

Note: This figure plots binned averages of the logarithm of one plus the migration rate against destination winter minimum temperature. Migration rates are defined as migration flows divided by origin population. The solid line represents a quadratic fit.

To examine whether migration responses change once temperature thresholds are crossed, we estimate a temperature-bin specification using pooled OLS with origin, destination, and year fixed effects. Instead of modeling temperature using quadratic terms, temperatures are grouped into bins representing extreme cold and extreme heat conditions. Moderate temperature ranges serve as the reference category.

This specification allows us to evaluate whether migration flows respond differently when provinces experience extreme climatic conditions. Temperatures below 0°C represent extremely cold winter conditions, while temperatures above 32°C represent extreme summer heat.

Table A.11 presents the results for the full sample of 81 provinces over the period 2008–2024. The estimates show that high temperatures in origin provinces increase migration flows, suggesting that extreme heat acts as a push factor. In contrast, extremely high temperatures in destination provinces reduce migration flows, indicating that migrants tend to avoid excessively hot destinations.

Table A.2: Three-Way Fixed Effects Estimates: Full Sample of 81 Provinces (Centered Kelvin Specification)

	Model 1	Model 2	Model 3	Model 4	Model 5
Origin Temperature (K)	3.983***	5.074***	0.860	0.375	0.391
	(0.84)	(0.89)	(0.99)	(0.98)	(0.98)
Origin Temperature^2		68.161***	115.304***	109.664***	109.548***
		(22.23)	(24.13)	(23.99)	(23.99)
Origin Precipitation (mm)			-0.046***	-0.050***	-0.050***
			(0.01)	(0.01)	(0.01)
Origin Precipitation^2			0.016***	0.012**	0.012**
			(0.01)	(0.01)	(0.01)
Wildfire Area (ha)			0.008***	0.008***	0.009***
			(0.00)	(0.00)	(0.00)
Electricity Consumption (kWh)				0.060***	0.061***
				(0.01)	(0.01)
Refugee Share (%)					0.006
					(0.02)
Constant	4.778***	4.764***	4.824***	4.355***	4.342***
	(0.01)	(0.01)	(0.01)	(0.10)	(0.12)
Observations	110,160	110,160	110,160	77,760	77,760
R-squared	0.741	0.741	0.850	0.862	0.862
Origin Fixed Effects	Yes	Yes	Yes	Yes	Yes
Destination Fixed Effects	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes
Control Variables	No	Yes	Yes	Yes	Yes

Note: The dependent variable is the logarithm of internal migration flows between provinces. This table reports the original nonlinear specification in which temperature is measured in Kelvin (K) and centered around its sample mean prior to estimation. Climate and economic variables enter the regressions in logarithmic form unless otherwise indicated, while squared terms capture potential nonlinear climate effects. The models follow a stepwise specification to evaluate the contribution of climate and socioeconomic factors to internal migration across the 81 provinces of Türkiye. Spatial distance is omitted because it is time-invariant and absorbed by the origin and destination fixed effects. This specification is retained as a robustness check; the preferred specification using the logarithm of temperature measured in degrees Celsius (°C) is reported in the main text. All regressions include origin, destination, and year fixed effects. Robust standard errors clustered at the origin–destination pair level are reported in parentheses. Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$.

Table A.3: PPML Estimates: Full Sample of 81 Provinces (Centered Kelvin Specification)

	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Origin Temperature (K)	4.277**	4.776**	5.034***	5.078***	4.611**	1.661
	(1.941)	(1.924)	(1.937)	(1.932)	(1.921)	(2.284)
Origin Temperature^2	178.338***	157.569***	158.847***	159.333***	163.866***	141.861***
	(45.158)	(45.093)	(44.615)	(44.325)	(44.111)	(45.507)
Origin Precipitation (mm)	-0.020*	-0.018	-0.018	-0.018	-0.016	-0.027**
	(0.012)	(0.012)	(0.012)	(0.012)	(0.012)	(0.013)
Origin GDP per pc ($)		-0.229***	-0.228***	-0.226***	-0.233***	-0.078*
		(0.054)	(0.054)	(0.053)	(0.054)	(0.046)
Destination Temperature (K)			6.783***	6.449***	5.928***	6.102***
			(1.599)	(1.618)	(1.599)	(1.905)
Destination Temperature^2			-45.351	-31.174	-48.033	-13.919
			(33.182)	(33.535)	(33.610)	(38.613)
Destination Precipitation (mm)			-0.019**	-0.021**	-0.019*	-0.031**
			(0.010)	(0.010)	(0.010)	(0.012)
Destination GDP pc ($)				0.164***	0.177***	0.187***
				(0.046)	(0.046)	(0.047)
Distance (km)					-0.764***	-0.768***
					(0.024)	(0.023)
Refugee Share (%)						0.025***
						(0.009)
Constant	7.002***	9.100***	9.197***	7.676***	12.409***	10.964***
	(0.022)	(0.491)	(0.487)	(0.642)	(0.628)	(0.644)
Observations	110,160	110,160	110,160	110,160	110,160	90,720
Pseudo R-squared	0.762	0.762	0.762	0.762	0.857	0.863
Origin Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Destination Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Control Variables	No	Yes	Yes	Yes	Yes	Yes

Note: The dependent variable is migration flows between origin and destination provinces and is estimated in levels using the PPML estimator. This table reports the original specification in which temperature is measured in Kelvin (K) and centered around its sample mean prior to estimation. Climate and economic variables are entered into the regressions in logarithmic form unless otherwise indicated. Climate variables include both origin and destination temperature and precipitation conditions, while GDP pc, distance, and refugee share are progressively introduced across specifications. This specification is retained as a robustness check; the preferred specification using the logarithm of temperature measured in (°C) is reported in the main text. All regressions include origin, destination, and year fixed effects. Robust standard errors clustered at the origin–destination pair level are reported in parentheses. Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$.

Table A.4: Hausman–Taylor Estimates: Full Sample of 81 Provinces (Centered Kelvin Specification)

	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Origin Temperature (K)	7.877***	9.369***	9.192***	6.967***	4.769***	3.963***
	(0.579)	(0.615)	(0.623)	(0.721)	(0.736)	(0.743)
Origin Temperature^2		103.6***	102.0***	179.9***	158.5***	149.3***
		(14.96)	(15.01)	(19.88)	(19.90)	(19.92)
Origin Precipitation (mm)			-0.314***	-0.245***	-0.280***	-0.230***
			(0.0656)	(0.0702)	(0.0701)	(0.0704)
Origin Precipitation^2			0.0245***	0.0166***	0.0189***	0.0146***
			(0.00519)	(0.00560)	(0.00559)	(0.00562)
Wildfire Area (ha)				0.00424***	0.00502***	0.00545***
				(0.000871)	(0.000871)	(0.000872)
Origin GDP pc ($)	-0.120***	-0.121***	-0.120***	-0.0757***	-0.0608***	-0.111***
	(0.0139)	(0.0140)	(0.0140)	(0.0157)	(0.0157)	(0.0161)
Refugee Share (%)					0.145***	0.138***
					(0.00994)	(0.00997)
Electricity Consumption (kWh)						0.0719***
						(0.00950)
Distance (km)	-0.709***	-0.712***	-0.712***	-0.727***	-0.704***	-0.702***
	(0.0253)	(0.0252)	(0.0253)	(0.0249)	(0.0251)	(0.0251)
Constant	10.19***	10.20***	11.19***	10.85***	10.68***	10.09***
	(0.208)	(0.208)	(0.292)	(0.306)	(0.308)	(0.311)
Observations	110,160	110,160	110,160	77,760	77,760	77,760
Pair Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes

Note: The dependent variable is the logarithm of internal migration flows between provinces. This table reports the original Hausman-Taylor specification in which temperature is measured in Kelvin (K) and centered around its sample mean prior to estimation. Climate and economic variables enter the regressions in logarithmic form unless otherwise indicated, while squared terms capture potential nonlinear climate effects. GDP pc is treated as an endogenous variable, whereas the remaining climate and socioeconomic variables are treated as exogenous. Geographic distance is identified through the internal instrumentation procedure of the Hausman-Taylor estimator. This specification is retained as a robustness check; the preferred specification using the logarithm of temperature measured in (°C) is reported in the main text. All specifications include province-pair and year fixed effects. Standard errors clustered at the province-pair level are reported in parentheses. Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$.

Table A.5: Three-Way Fixed Effects Estimates: Climate-Exposed Subsample of 30 Provinces (Centered Kelvin Specification)

	Model 1	Model 2	Model 3	Model 4	Model 5
Origin Temperature (K)	10.421***	10.965***	13.724***	13.322***	13.762***
	(1.76)	(1.72)	(1.82)	(1.80)	(1.81)
Origin Temperature^2		50.405	69.836	63.011	56.810
		(52.92)	(57.56)	(56.62)	(56.90)
Origin Precipitation (mm)		-0.051***	-0.105***	-0.108***	-0.107***
		(0.01)	(0.01)	(0.01)	(0.01)
Origin Precipitation^2		0.007	0.015*	0.016**	0.017**
		(0.01)	(0.01)	(0.01)	(0.01)
Wildfire Area (ha)			0.010***	0.011***	0.011***
			(0.00)	(0.00)	(0.00)
Electricity Consumption (kWh)				0.021	0.022
				(0.02)	(0.02)
Refugee Share (%)					0.036
					(0.04)
Constant	4.678***	4.669***	4.731***	4.576***	4.477***
	(0.01)	(0.01)	(0.02)	(0.14)	(0.17)
Observations	40,800	40,800	40,800	28,800	28,800
R-squared	0.725	0.725	0.853	0.866	0.866
Origin Fixed Effects	Yes	Yes	Yes	Yes	Yes
Destination Fixed Effects	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes
Control Variables	No	No	Yes	Yes	Yes

Note: The dependent variable is the logarithm of internal migration flows between provinces. This table reports the original nonlinear specification in which temperature is measured in Kelvin (K) and centered around its sample mean prior to estimation. Climate and economic variables enter the regressions in logarithmic form unless otherwise indicated, while squared terms capture potential nonlinear climate effects. The sample is restricted to 30 origin provinces identified as highly exposed to climatic stress, while all 81 provinces remain potential destinations. Spatial distance is omitted because it is time-invariant and absorbed by the origin and destination fixed effects. This specification is retained as a robustness check; the preferred specification using the logarithm of temperature measured in (°C) is reported in the main text. All regressions include origin, destination, and year fixed effects. Robust standard errors clustered at the origin–destination pair level are reported in parentheses. Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$.

Table A.6: PPML Estimates: Climate-Exposed Subsample of 30 Provinces (Centered Kelvin Specification)

	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Origin Temperature (K)	9.555***	11.117***	11.261***	11.239***	10.509***	8.479**
	(3.626)	(3.586)	(3.580)	(3.572)	(3.528)	(3.689)
Origin Temperature^2	175.401*	155.855	155.097	154.959	173.679*	277.373**
	(102.208)	(102.780)	(101.826)	(101.293)	(99.725)	(114.127)
Origin Precipitation (mm)	-0.122***	-0.115***	-0.115***	-0.115***	-0.112***	-0.157***
	(0.026)	(0.025)	(0.025)	(0.025)	(0.025)	(0.031)
Origin GDP pc ($)		-0.262***	-0.262***	-0.258***	-0.266***	-0.092
		(0.080)	(0.079)	(0.078)	(0.078)	(0.061)
Destination Temperature (K)			12.147***	11.262***	9.987***	10.080***
			(2.903)	(2.814)	(2.729)	(3.689)
Destination Temperature^2			-112.859	-93.875	-109.709	-41.168
			(69.244)	(69.734)	(68.629)	(74.781)
Destination Precipitation (mm)			-0.011	-0.015	-0.002	-0.016
			(0.018)	(0.018)	(0.018)	(0.023)
Destination GDP pc ($)				0.269***	0.285***	0.218***
				(0.066)	(0.067)	(0.072)
Distance (km)					-1.141***	-1.129***
					(0.041)	(0.040)
Refugee Share (%)						0.029**
						(0.015)
Constant	6.606***	8.900***	8.923***	6.441***	13.593***	12.643***
	(0.036)	(0.695)	(0.681)	(0.902)	(0.884)	(0.929)
Observations	40,800	40,800	40,800	40,800	40,800	33,600
Pseudo R-squared	0.736	0.736	0.736	0.737	0.873	0.876
Origin Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Destination Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Control Variables	No	Yes	Yes	Yes	Yes	Yes

Note: The dependent variable is migration flows between origin and destination provinces and is estimated in levels using the PPML estimator. This table reports the original specification in which temperature is measured in Kelvin (K) and centered around its sample mean prior to estimation. Climate and economic variables are entered into the regressions in logarithmic form unless otherwise indicated. Climate variables include both origin and destination temperature and precipitation conditions, while GDP pc, distance, and refugee share are progressively introduced across specifications. The sample is restricted to the 30 origin provinces identified as highly exposed to climatic stress, while all 81 provinces remain potential destinations. This specification is retained as a robustness check; the preferred

Table A.7: Hausman–Taylor Estimates: Climate-Exposed Subsample of 30 Provinces (Centered Kelvin Specification)

	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Origin Temperature (K)	16.73***	17.54***	16.84***	19.87***	19.83***	19.49***
	(1.088)	(1.114)	(1.125)	(1.368)	(1.437)	(1.454)
Origin Temperature^2		101.0***	83.55***	135.6***	135.0***	131.3***
		(30.44)	(30.68)	(43.94)	(44.39)	(44.53)
Origin Precipitation (mm)			-0.103	-0.0221	-0.0221	-0.0439
			(0.104)	(0.109)	(0.109)	(0.110)
Origin Precipitation^2			0.00521	0.00870	0.00870	0.0106
			(0.00826)	(0.00877)	(0.00877)	(0.00883)
Wildfire Area (ha)				0.00844***	0.00844***	0.00865***
				(0.00160)	(0.00160)	(0.00160)
Origin GDP pc ($)	-0.243***	-0.241***	-0.237***	-0.0720***	-0.0697***	-0.110***
	(0.0209)	(0.0209)	(0.0209)	(0.0239)	(0.0243)	(0.0249)
Refugee Share (%)					0.00125	-0.00391
					(0.0143)	(0.0143)
Electricity Consumption (kWh)						0.0253*
						(0.0137)
Distance (km)	-0.602***	-0.603***	-0.600***	-0.616***	-0.616***	-0.615***
	(0.0403)	(0.0402)	(0.0404)	(0.0393)	(0.0393)	(0.0394)
Constant	10.39***	10.36***	10.74***	9.553***	9.529***	9.772***
	(0.322)	(0.322)	(0.459)	(0.483)	(0.489)	(0.492)
Observations	40,800	40,800	40,800	28,800	28,800	28,800
Province Pairs	2,400	2,400	2,400	2,400	2,400	2,400
Pair Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes

Note: The dependent variable is the logarithm of internal migration flows between provinces. This table reports the original Hausman–Taylor specification in which temperature is measured in Kelvin (K) and centered around its sample mean prior to estimation. Climate and economic variables enter the regressions in logarithmic form unless otherwise indicated, while squared terms capture potential nonlinear climate effects. The sample is restricted to the 30 origin provinces identified as highly exposed to climatic stress, while all 81 provinces remain potential destinations. Origin GDP per capita is treated as endogenous, whereas the remaining climate and socioeconomic variables are treated as exogenous. Geographic distance is identified through the internal instrumentation procedure of the Hausman–Taylor estimator. This specification is retained as a robustness check; the preferred specification using the logarithm of temperature measured in (°C) is reported in the main text. All specifications include province–pair and year fixed effects. Standard errors clustered at the province–pair level are reported in parentheses. Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$.

Table A.8: Three-Way Fixed Effects Estimates: Drought-Exposed Subsample of 17 Provinces (Centered Kelvin Specification)

	Model 1	Model 2	Model 3	Model 4	Model 5
Origin Temperature (K)	0.572	1.477	1.973	1.000	6.616**
	(2.27)	(2.18)	(2.61)	(2.60)	(2.57)
Origin Temperature^2		255.341***	221.364**	207.128**	98.964
		(76.37)	(89.30)	(88.65)	(89.18)
Origin Precipitation (mm)		-0.063***	-0.110***	-0.116***	-0.113***
		(0.01)	(0.02)	(0.02)	(0.02)
Origin Precipitation^2		0.022**	0.023**	0.026**	0.028***
		(0.01)	(0.01)	(0.01)	(0.01)
Wildfire Area (ha)			0.008	0.007	0.008
			(0.01)	(0.01)	(0.01)
Electricity Consumption (kWh)				0.039*	0.059**
				(0.02)	(0.02)
Refugee Share (%)					0.486***
					(0.05)
Constant	4.802***	4.758***	4.849***	4.567***	2.874***
	(0.02)	(0.02)	(0.03)	(0.18)	(0.23)
Observations	23,120	23,120	23,120	16,320	16,320
R-squared	0.819	0.819	0.884	0.896	0.896
Origin Fixed Effects	Yes	Yes	Yes	Yes	Yes
Destination Fixed Effects	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes
Control Variables	No	Yes	Yes	Yes	Yes

Note: The dependent variable is the logarithm of internal migration flows between provinces. This table reports the original nonlinear specification in which temperature is measured in Kelvin (K) and centered around its sample mean prior to estimation. Climate and economic variables enter the regressions in logarithmic form unless otherwise indicated, while squared terms capture potential nonlinear climate effects. The sample is restricted to 17 origin provinces identified as highly exposed to drought conditions, while all 81 provinces remain potential destinations. Spatial distance is omitted because it is time-invariant and absorbed by the origin and destination fixed effects. This specification is retained as a robustness check; the preferred specification using the logarithm of temperature measured in (°C) is reported in the main text. All regressions include origin, destination, and year fixed effects. Robust standard errors clustered at the origin–destination pair level are reported in parentheses. Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$.

Table A.9: PPML Estimates: Drought-Exposed Subsample of 17 Provinces (Centered Kelvin Specification)

	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Origin Temperature (K)	49.225***	55.733***	54.443***	50.908***	44.082***	31.440***
	(2.736)	(3.165)	(3.174)	(2.901)	(3.001)	(4.696)
Origin Temperature^2	179.283	194.139	303.056	198.046	221.897	83.444
	(202.352)	(220.121)	(213.168)	(198.219)	(200.242)	(244.646)
Origin Precipitation (mm)	-0.027	-0.094***	-0.059*	-0.001	-0.155***	-0.208***
	(0.035)	(0.035)	(0.034)	(0.034)	(0.033)	(0.038)
Origin GDP pc ($)		0.415***	0.372***	0.151***	0.304***	0.239***
		(0.041)	(0.040)	(0.041)	(0.046)	(0.049)
Destination Temperature (K)			47.127***	44.720***	33.676***	32.771***
			(0.865)	(0.985)	(0.950)	(1.001)
Destination Temperature^2			-236.137***	-661.092***	-481.836***	-469.911***
			(66.287)	(55.161)	(56.049)	(60.428)
Destination Precipitation (mm)			-0.126***	-0.247***	-0.031	-0.036
			(0.032)	(0.032)	(0.031)	(0.033)
Destination GDP pc ($)				0.991***	2.009***	1.930***
				(0.055)	(0.061)	(0.067)
Distance (km)					-1.009***	-0.983***
					(0.019)	(0.021)
Refugee Share (%)						0.052***
						(0.010)
Constant	5.404***	1.780***	2.826***	-3.476***	-3.530***	-3.927***
	(0.023)	(0.360)	(0.389)	(0.608)	(0.533)	(0.591)
Observations	23,120	23,120	23,120	23,120	23,120	19,040
Pseudo R-squared	0.791	0.791	0.792	0.792	0.884	0.888
Origin Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Destination Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Control Variables	No	Yes	Yes	Yes	Yes	Yes

Note: The dependent variable is migration flows between origin and destination provinces and is estimated in levels using the PPML estimator. This table reports the original specification in which temperature is measured in Kelvin (K) and centered around its sample mean prior to estimation. Climate and economic variables are entered into the regressions in logarithmic form unless otherwise indicated. Climate variables include both origin and destination temperature and precipitation conditions, while GDP pc, distance, and refugee share are progressively introduced across specifications. The sample is restricted to the 17 origin provinces identified as highly exposed to drought conditions, while all 81 provinces remain potential destinations. This specification is retained as a robustness check; the preferred specification using the logarithm of temperature measured in (°C) is reported in the main text. All regressions include origin, destination, and year fixed effects. Robust standard errors clustered at the origin–destination pair level are reported in parentheses. Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$.

Table A.10: Hausman–Taylor Estimates: Drought-Exposed Subsample of 17 Provinces (Centered Kelvin Specification)

	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Origin Temperature (K)	9.373***	9.884***	9.307***	13.66***	9.919***	8.621***
	(1.441)	(1.442)	(1.450)	(2.169)	(2.232)	(2.262)
Origin Temperature^2		106.9***	101.4***	136.2***	134.7***	129.5***
		(44.60)	(44.68)	(62.33)	(62.77)	(63.07)
Origin Precipitation (mm)			-0.320	-0.165	-0.197	-0.126
			(0.217)	(0.231)	(0.231)	(0.233)
Origin Precipitation^2			0.0210	0.0101	0.0125	0.0076
			(0.0163)	(0.0177)	(0.0177)	(0.0179)
Wildfire Area (ha)				0.00829***	0.00740***	0.00750***
				(0.00229)	(0.00227)	(0.00227)
Origin GDP pc ($)	-0.342***	-0.341***	-0.342***	-0.176***	-0.106***	-0.179***
	(0.0312)	(0.0313)	(0.0313)	(0.0360)	(0.0372)	(0.0381)
Refugee Share (%)					0.121***	0.119***
					(0.0212)	(0.0213)
Electricity Consumption (kWh)						0.0575***
						(0.0171)
Distance (km)	-0.669***	-0.670***	-0.669***	-0.697***	-0.692***	-0.691***
	(0.0558)	(0.0558)	(0.0558)	(0.0543)	(0.0543)	(0.0544)
Constant	11.40***	11.10***	12.04***	10.97***	10.09***	9.652***
	(0.424)	(0.425)	(0.590)	(0.625)	(0.666)	(0.671)
Observations	23,120	23,120	23,120	16,320	16,320	16,320
Pair Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes

Note: The dependent variable is the logarithm of internal migration flows between provinces. This table reports the original Hausman–Taylor specification in which temperature is measured in Kelvin (K) and centered around its sample mean prior to estimation. Climate and economic variables enter the regressions in logarithmic form unless otherwise indicated, while squared terms capture potential nonlinear climate effects. The sample is restricted to the 17 origin provinces identified as highly exposed to drought conditions, while all 81 provinces remain potential destinations. Origin GDP per capita is treated as endogenous, whereas the remaining climate and socioeconomic variables are treated as exogenous. Geographic distance is identified through the internal instrumentation procedure of the Hausman–Taylor estimator. This specification is retained as a robustness check; the preferred specification using the logarithm of temperature measured in (°C) is reported in the main text. All specifications include province–pair and year fixed effects. Standard errors clustered at the province–pair level are reported in parentheses. Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$.

Table A.11: Temperature Bin Results for Internal Migration: 81 Provinces in Türkiye (2008–2024)

	Model 1	Model 2	Model 3	Model 4
Origin Temperature				
Average Temperature (K)	4.309***	3.983***	4.128***	–
	(0.817)	(0.842)	(0.838)	
Cold Bin (<0°C)	–	–	–	-0.019***
				(0.005)
Hot Bin (>32°C)	–	–	–	0.014**
				(0.006)
Destination Temperature				
Average Temperature (K)	–	–	11.660***	–
			(0.891)	
Cold Bin (<0°C)	–	–	–	-0.005
				(0.004)
Hot Bin (>32°C)	–	–	–	-0.012**
				(0.006)
Control Variables				
Distance (km)	-0.787***	-0.787***	-0.787***	-0.787***
	(0.014)	(0.014)	(0.014)	(0.014)
Origin Precipitation (mm)	–	-0.015***	-0.015***	-0.020***
		(0.005)	(0.005)	(0.005)
Destination Precipitation (mm)	–	–	0.013***	0.002
			(0.005)	(0.005)
Origin Fixed Effects	Yes	Yes	Yes	Yes
Destination Fixed Effects	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes
Observations	110,160	110,160	110,160	110,160
Moderate Temperature Bin (Reference)	–	–	–	Yes

Note: The dependent variable is the logarithm of one plus internal migration flows between provinces. Models 1–3 measure temperature continuously in Kelvin (K), whereas Model 4 replaces the continuous temperature terms with discrete temperature bins. The cold bin denotes province-years with average temperatures below 0°C, and the hot bin denotes those above 32°C; the moderate temperature bin serves as the omitted reference category. Precipitation is measured in millimeters (mm) and distance in kilometers (km). All specifications include fixed effects for origin, destination, and year. Robust standard errors clustered at the origin–destination pair level are reported in parentheses. Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$.

Table A.1: Provinces Included in the Climate-Exposed and Drought-Exposed Subsamples

Climate-Exposed Provinces (30)	**Drought-Exposed Provinces (17)**
Adıyaman	Adıyaman
Aksaray	Aksaray
Antalya	Antalya
Batman	Batman
Bayburt	Bingöl
Bingöl	Diyarbakır
Diyarbakır	Elazığ
Elazığ	Gaziantep
Erzincan	Hakkâri
Gaziantep	Hatay
Gümüşhane	Kahramanmaraş
Hakkâri	Karaman
Hatay	Kilis
Kahramanmaraş	Mardin
Karaman	Şanlıurfa
Kastamonu	Siirt
Kilis	Şırnak
Kırşehir	
Mardin	
Muğla	
Niğde	
Ordu	
Osmaniye	
Rize	
Şanlıurfa	
Siirt	
Şırnak	
Tokat	
Van	
Yozgat	

Note: The climate-exposed subsample consists of 30 provinces selected based on climatic exposure, geographic location, and urban–rural characteristics. The drought-exposed subsample consists of 17 provinces identified using the Bagnouls–Gaussen aridity index following Cebeci et al. (2019). These subsamples are used to examine whether climate-related migration responses are stronger in regions that are more vulnerable to environmental stress.

References

Afifi, Tamer and Koko Warner (2008). "The Impact of Environmental Degradation on Migration Flows across Countries". In: *UNU-EHS Working Paper*.

Anderson, James E. and Eric van Wincoop (2003). "Gravity with Gravitas: A Solution to the Border Puzzle". In: *American Economic Review* 93.1, pp. 170–192. DOI: 10.1257/000282803321455214.

Anderson, James E. and Yoto V. Yotov (2025). "Gravity in Transition". In: *Review of International Economics*. DOI: 10.1111/roie.12799.

Backhaus, Andreas, Inmaculada Martinez-Zarzoso, and Chris Muris (2015). "Do Climate Variations Explain Bilateral Migration? A Gravity Model Analysis". In: *IZA Journal of Migration* 4.1, p. 3. DOI: 10.1186/s40176-014-0026-3.

Baltagi, Badi H., Georges Bresson, and Alain Pirotte (2003). "Fixed Effects, Random Effects or Hausman–Taylor?: A Pretest Estimator". In: *Economics Letters* 79.3, pp. 361–369. DOI: 10.1016/S0165-1765(03)00007-7.

Barrios, Salvador, Luisito Bertinelli, and Eric Strobl (2006). "Climatic Change and Rural–Urban Migration: The Case of Sub-Saharan Africa". In: *Journal of Urban Economics* 60.3, pp. 357–371. DOI: 10.1016/j.jue.2006.04.005.

Bayram, Hasan and Ayşe Bilge Öztürk (2021). "Global Climate Change, Desertification, and Its Consequences in Turkey and the Middle East". In: *Climate Change and Global Public Health*. Cham: Springer, pp. 445–458. DOI: 10.1007/978-3-030-54746-2_21.

Becker, Gary S. (1962). "Investment in Human Capital: A Theoretical Analysis". In: *Journal of Political Economy* 70.5, pp. 9–49. DOI: 10.1086/258724.

Beine, Michel, Simone Bertoli, and Jesús Fernández-Huertas Moraga (2016). "A Practitioners' Guide to Gravity Models of International Migration". In: *The World Economy* 39.4, pp. 496–512. DOI: 10.1111/twec.12265.

Beine, Michel, Frédéric Docquier, and Çağlar Özden (2011). "Diasporas". In: *Journal of Development Economics* 95.1, pp. 30–41. DOI: 10.1016/j.jdeveco.2009.11.004.

Beine, Michel and Lionel Jeusette (2021). "A Meta-Analysis of the Literature on Climate Change and Migration". In: *Journal of Demographic Economics* 87.3, pp. 293–344. DOI: 10.1017/dem.2019.22.

Beine, Michel and Christopher Parsons (2015). “Climatic Factors as Determinants of International Migration”. In: *Scandinavian Journal of Economics* 117.2, pp. 723–767. DOI: 10.1111/sjoe.12098.

Beine, Michel and Christopher R. Parsons (2017). “Climatic Factors as Determinants of International Migration: Redux”. In: *CESifo Economic Studies* 63.4, pp. 386–402. DOI: 10.1093/cesifo/ifx017.

Benli, H. et al. (2018). *Türkiye’de Afet Yönetimi ve Doğa Kaynaklı Afet İstatistikleri*. Ankara: Afet ve Acil Durum Yönetimi Başkanlığı (AFAD).

Benveniste, Hélène, Michael Oppenheimer, and Marc Fleurbaey (2022). “Climate change increases resource-constrained international immobility”. In: *Nature Climate Change* 12.7, pp. 634–641. DOI: 10.1038/s41558-022-01401-w. URL: https://doi.org/10.1038/s41558-022-01401-w.

Berlemann, Michael and Max Friedrich Steinhardt (2017). “Climate Change, Natural Disasters, and Migration—A Survey of the Empirical Evidence”. In: *CESifo Economic Studies* 63.4, pp. 353–385. DOI: 10.1093/cesifo/ifx019.

Berlemann, Michael and Thi Xuyen Tran (2020). “Climate-Related Hazards and Internal Migration: Empirical Evidence for Rural Vietnam”. In: *Economics of Disasters and Climate Change* 4.3, pp. 385–409. DOI: 10.1007/s41885-020-00062-3.

Black, Richard (2001). *Environmental Refugees: Myth or Reality?* Tech. rep. 34. Accessed: 17 April 2026. United Nations High Commissioner for Refugees (UNHCR). URL: https://www.refworld.org/reference/research/unhcr/2001/87301.

Black, Richard et al. (2011). “The Effect of Environmental Change on Human Migration”. In: *Global Environmental Change* 21.S1, S3–S11. DOI: 10.1016/j.gloenvcha.2011.10.001.

Bohra-Mishra, Pratikshya, Michael Oppenheimer, and Solomon M. Hsiang (2014). “Nonlinear Permanent Migration Response to Climatic Variations but Minimal Response to Disasters”. In: *Proceedings of the National Academy of Sciences* 111.27, pp. 9780–9785. DOI: 10.1073/pnas.1317166111.

Bozkurt, Deniz and Omer Lutfi Sen (2013). “Climate Change Impacts in the Euphrates–Tigris Basin Based on Different Model and Scenario Simulations”. In: *Journal of Hydrology* 480, pp. 149–161. DOI: 10.1016/j.jhydrol.2012.12.021.

Brown, Oli (2008). *Migration and Climate Change*. IOM Migration Research Series. Geneva: International Organization for Migration. DOI: 10.18356/26de4416-en.

Burke, Marshall, Solomon M. Hsiang, and Edward Miguel (2015). “Global Non-Linear Effect of Temperature on Economic Production”. In: *Nature* 527.7577, pp. 235–239. DOI: 10.1038/nature15725.

Burzyński, Michał et al. (2022). “Climate Change, Inequality, and Human Migration”. In: *Journal of the European Economic Association* 20.3, pp. 1145–1197. DOI: 10.1093/jeea/jvab054.

Cai, Ruohong et al. (2016). “Climate Variability and International Migration: The Importance of the Agricultural Linkage”. In: *Journal of Environmental Economics and Management* 79, pp. 135–151. DOI: 10.1016/j.jeem.2016.06.005.

Campbell, John and Oliver Warrick (2014). *Climate Change and Migration Issues in the Pacific*. New York: United Nations Economic, Social Commission for Asia, and the Pacific (ESCAP).

Carleton, Tamma, Solomon M. Hsiang, and Marshall Burke (2016). “Conflict in a Changing Climate”. In: *European Physical Journal Special Topics* 225.3, pp. 489–511. DOI: 10.1140/epjst/e2015-50100-5.

Carleton, Tamma, Amir Jina, et al. (2022). “Valuing the Global Mortality Consequences of Climate Change Accounting for Adaptation Costs and Benefits”. In: *The Quarterly Journal of Economics* 137.4, pp. 2037–2105. DOI: 10.1093/qje/qjac020.

Cattaneo, Cristina, Michel Beine, et al. (2019). “Human Migration in the Era of Climate Change”. In: *Review of Environmental Economics and Policy* 13.2, pp. 189–206. DOI: 10.1093/reep/rez008.

Cattaneo, Cristina and Giovanni Peri (2016). “The Migration Response to Increasing Temperatures”. In: *Journal of Development Economics* 122, pp. 127–146. DOI: 10.1016/j.jdeveco.2016.05.004.

Cebeci, İlknur et al. (2019). “Türkiye’nin İller Bazında Kuraklık Değerlendirmesi”. In: *Toprak Su Dergisi*, pp. 169–176. DOI: 10.21657/topraksu.655613.

Chen, Joyce and Valerie Mueller (2019). “Climate-Induced Cross-Border Migration and Change in Demographic Structure”. In: *Population and Environment* 41.2, pp. 98–125. DOI: 10.1007/s11111-019-00328-3.

Cifci, Eren and Matthew E. Oliver (2018). “Reassessing the Links Between GHG Emissions, Economic Growth, and the UNFCCC: A Difference-in-Differences Approach”. In: *Sustainability* 10.2, p. 334. DOI: 10.3390/su10020334.

Coniglio, Nicola D. and Giovanni Pesce (2015). “Climate Variability and International Migration: An Empirical Analysis”. In: *Environment and Development Economics* 20.4, pp. 434–468. DOI: 10.1017/S1355770X14000722.

Dallmann, Ingrid and Katrin Millock (2017). "Climate Variability and Inter-State Migration in India". In: *CESifo Economic Studies* 63.4, pp. 560–594. DOI: 10.1093/cesifo/ifx014.

Dell, Melissa, Benjamin F. Jones, and Benjamin A. Olken (2014). "What Do We Learn from the Weather? The New Climate-Economy Literature". In: *Journal of Economic Literature* 52.3, pp. 740–798. DOI: 10.1257/jel.52.3.740.

Demircan, Mesut et al. (2017). "Climate Change Projections for Turkey: Three Models and Two Scenarios". In: *Turkish Journal of Water Science and Management* 1.1, pp. 22–43. DOI: 10.31807/tjwsm.297183.

Egger, Peter and Michael Pfaffermayr (2004). "Distance, Trade and FDI: A Hausman–Taylor SUR Approach". In: *Journal of Applied Econometrics* 19.2, pp. 227–246. DOI: 10.1002/jae.721.

Erdoğan, Zahide and Safure Cantürk (2022). "Understanding the Climate-Conflict-Migration Nexus: Immigration from Climate-Conflict Zones to Turkey". In: *SİYASAL: Journal of Political Sciences* 31.1, pp. 137–155. DOI: 10.26650/siyasal.2022.31.994670.

Freeman, Rebecca and John Lewis (2021). "Gravity Model Estimates of the Spatial Determinants of Trade, Migration, and Trade-and-Migration Policies". In: *Economics Letters* 204, p. 109873. DOI: 10.1016/j.econlet.2021.109873.

Friedman, Milton (1957). "The Permanent Income Hypothesis". In: *A Theory of the Consumption Function*. Princeton, NJ: Princeton University Press, pp. 20–37.

Gray, Clark, Douglas Hopping, and Valerie Mueller (2020). "The Changing Climate-Migration Relationship in China, 1989–2011". In: *Climatic Change* 160, pp. 103–122. DOI: 10.1007/s10584-020-02657-x.

Gray, Clark L. (2009). "Environment, Land, and Rural Out-Migration in the Southern Ecuadorian Andes". In: *World Development* 37.2, pp. 457–468. DOI: 10.1016/j.worlddev.2008.05.004.

Grohmann, Tobias (2023). *Cultural Similarity and Migration: New Evidence from a Gravity Model of International Migration*. IOS Working Papers 404. Regensburg: Leibniz Institute for East and Southeast European Studies (IOS). URL: https://hdl.handle.net/10419/281137.

Halliday, Timothy (2006). "Migration, Risk, and Liquidity Constraints in El Salvador". In: *Economic Development and Cultural Change* 54.4, pp. 893–925. DOI: 10.1086/503584.

Hausman, Jerry A. and William E. Taylor (1981). "Panel Data and Unobservable Individual Effects". In: *Econometrica* 49.6, pp. 1377–1398. DOI: 10.2307/1911406.

Helbling, Marc and Daniel Meierrieks (2021). "How Climate Change Leads to Emigration: Conditional and Long-Run Effects". In: *Review of Development Economics* 25.4, pp. 2323–2349. DOI: 10.1111/rode.12800.

El-Hinnawi, Essam (1985). *Environmental Refugees*. Nairobi: United Nations Environment Programme (UNEP).

Hsiao, Allan (2026). "Coordination and Commitment in International Climate Action: Evidence from Palm Oil". In: *Econometrica*. DOI: 10.3982/ECTA20608.

IPCC (2022). "Summary for Policymakers". In: *Climate Change 2022: Impacts, Adaptation, and Vulnerability. Contribution of Working Group II to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*. Ed. by H.-O. Poertner et al. Cambridge, UK and New York, USA: Cambridge University Press, pp. 3–33.

Joseph, George and Quentin Wodon (2013). "Is Internal Migration in Yemen Driven by Climate or Socio-Economic Factors?" In: *Review of International Economics* 21.2, pp. 295–310. DOI: 10.1111/roie.12037.

Kelley, Colin et al. (2017). "Commentary on the Syria Case: Climate as a Contributing Factor". In: *Political Geography* 60, pp. 245–247. DOI: 10.1016/j.polgeo.2017.06.013.

Kharin, Viatcheslav V. et al. (2007). "Changes in Temperature and Precipitation Extremes in the IPCC Ensemble of Global Coupled Model Simulations". In: *Journal of Climate* 20.8, pp. 1419–1444. DOI: 10.1175/JCLI4066.1.

Koubi, Vally, Sebastian Stoll, and Gabriele Spilker (2016). "Perceptions of Environmental Change and Migration Decisions". In: *Climatic Change* 138, pp. 439–451. DOI: 10.1007/s10584-016-1767-1.

Krishnamurthy, P. Krishna, Kirsty Lewis, and R. J. Choularton (2014). "A Methodological Framework for Rapidly Assessing the Impacts of Climate Risk on National-Level Food Security through a Vulnerability Index". In: *Global Environmental Change* 25, pp. 121–132. DOI: 10.1016/j.gloenvcha.2013.11.004.

Linders, Gert-Jan M. and Henri L. F. de Groot (2006). *Estimation of the Gravity Equation in the Presence of Zero Flows*. Tinbergen Institute Discussion Paper 06-072/3. Amsterdam and Rotterdam: Tinbergen Institute. URL: https://hdl.handle.net/10419/86589.

McCallum, John (1995). "National Borders Matter: Canada–US Regional Trade Patterns". In: *American Economic Review* 85.3, pp. 615–623.

Millock, Katrin (2015). "Migration and Environment". In: *Annual Review of Resource Economics* 7.1, pp. 35–60. DOI: 10.1146/annurev-resource-100814-125031.

Missirian, Anouch and Wolfram Schlenker (2017). “Asylum Applications and Migration Flows”. In: *American Economic Review* 107.5, pp. 436–440. DOI: 10.1257/aer.p20171051.

Moore, Maya and Dennis Wesselbaum (2023). “Climatic Factors as Drivers of Migration: A Review”. In: *Environment, Development and Sustainability* 25.4, pp. 2955–2975. DOI: 10.1007/s10668-022-02191-z.

Munshi, Kaivan (2003). “Networks in the Modern Economy: Mexican Migrants in the US Labor Market”. In: *Quarterly Journal of Economics* 118.2, pp. 549–599. DOI: 10.1162/003355303321675455.

Murray-Tortarolo, Guillermo N. and Mario Martínez Salgado (2021). “Drought as a Driver of Mexico–US Migration”. In: *Climatic Change* 164, p. 48. DOI: 10.1007/s10584-021-03030-2.

Myers, Norman (2002). “Environmental Refugees: A Growing Phenomenon of the 21st Century”. In: *Philosophical Transactions of the Royal Society B* 357.1420, pp. 609–613. DOI: 10.1098/rstb.2001.0953.

Nawrotzki, Raphael J. et al. (2015). “Amplification or Suppression: Social Networks and the Climate Change–Migration Association in Rural Mexico”. In: *Global Environmental Change* 35, pp. 463–474. DOI: 10.1016/j.gloenvcha.2015.09.002.

Nordhaus, William D. (2013). *The Climate Casino: Risk, Uncertainty, and Economics for a Warming World*. New Haven, CT: Yale University Press.

— (2019). “Climate Change: The Ultimate Challenge for Economics”. In: *American Economic Review* 109.6, pp. 1991–2014. DOI: 10.1257/aer.109.6.1991.

Nordhaus, William D. and Joseph Boyer (2003). *Warming the World: Economic Models of Global Warming*. Cambridge, MA: MIT Press.

Ozceylan, Dilek and Erman Coskun (2012). “Turkiye’deki Illerin Sosyo-Ekonomik Gelismislik Duzeyleri ve Afetlerden Sosyal ve Ekonomik Zarar Gorebilirlikleri Arasindaki Iliski”. In: *Istanbul University Journal of the School of Business Administration* 41.1, pp. 31–46.

Piguet, Etienne, Raoul Kaenzig, and Jeremie Guelat (2018). “The Uneven Geography of Research on Environmental Migration”. In: *Population and Environment* 39.4, pp. 357–383. DOI: 10.1007/s11111-018-0296-4.

Piguet, Etienne, Antoine Pecoud, and Paul De Guchteneire (2011). “Migration and Climate Change: An Overview”. In: *Refugee Survey Quarterly* 30.3, pp. 1–23. DOI: 10.1093/rsq/hdr006.

Rapoport, Hillel, Sulin Sardoschau, and Arthur Silve (2022). *Migration and Cultural Change*. Working Paper. Paris School of Economics and Humboldt University.

Rappaport, Jordan (2009). “Moving to Nice Weather”. In: *Environmental Amenities and Regional Economic Development*. Ed. by Todd L. Cherry and Dan S. Rickman. London: Routledge, pp. 25–53. DOI: 10.4324/9780203868485.

Ravenstein, E. G. (1885). “The Laws of Migration”. In: *Journal of the Royal Statistical Society* 48.2, pp. 167–235. DOI: 10.2307/2979181.

Reuveny, Rafael and Will H. Moore (2009). “Does Environmental Degradation Influence Migration? Emigration to Developed Countries in the Late 1980s and 1990s”. In: *Social Science Quarterly* 90.3, pp. 461–479. DOI: 10.1111/j.1540-6237.2009.00569.x.

Rigaud, Kanta Kumari et al. (2018). *Groundswell: Preparing for Internal Climate Migration*. Tech. rep. Washington, DC: World Bank. DOI: 10.1596/29461.

Roberts, Michael J., Wolfram Schlenker, and Jonathan Eyer (2013). “Agronomic Weather Measures in Econometric Models of Crop Yield with Implications for Climate Change”. In: *American Journal of Agricultural Economics* 95.2, pp. 236–243. DOI: 10.1093/ajae/aas047.

Ruiz, Vicente (2017). “Do Climatic Events Influence Internal Migration? Evidence from Mexico”. In: *FAERE Working Paper* 2017.19, pp. 1–19.

Rummukainen, Markku (2012). “Changes in Climate and Weather Extremes in the 21st Century”. In: *Wiley Interdisciplinary Reviews: Climate Change* 3.2, pp. 115–129. DOI: 10.1002/wcc.160.

Santana-Gallego, Maria and Jordi Paniagua (2022). “Tourism and Migration: Identifying the Channels with Gravity Models”. In: *Tourism Economics* 28.2, pp. 394–417. DOI: 10.1177/1354816620972806.

Santos Silva, J. M. C. and Silvana Tenreyro (2006). “The Log of Gravity”. In: *The Review of Economics and Statistics* 88.4, pp. 641–658. DOI: 10.1162/rest.88.4.641.

— (2011). “Further Simulation Evidence on the Performance of the Poisson Pseudo-Maximum Likelihood Estimator”. In: *Economics Letters* 112.2, pp. 220–222. DOI: 10.1016/j.econlet.2011.05.008.

Sasser, Alicia C. (2010). “Voting with Their Feet: Relative Economic Conditions and State Migration Patterns”. In: *Regional Science and Urban Economics* 40.2–3, pp. 122–135. DOI: 10.1016/j.regsciurbeco.2010.02.001.

Schlenker, Wolfram and Michael J. Roberts (2009). “Nonlinear Temperature Effects Indicate Severe Damages to U.S. Crop Yields Under Climate Change”. In: *Proceedings of the National Academy of Sciences* 106.37, pp. 15594–15598. DOI: 10.1073/pnas.0906865106.

Selby, Jan et al. (2017). “Climate Change and the Syrian Civil War Revisited”. In: *Political Geography* 60, pp. 232–244. DOI: 10.1016/j.polgeo.2017.05.007.

Serkendiz, Hıdır et al. (2023). “Multidimensional Assessment of Agricultural Drought Vulnerability Based on Socioeconomic and Biophysical Indicators”. In: *International Journal of Disaster Risk Reduction* 98, p. 104121. DOI: 10.1016/j.ijdrr.2023.104121.

Serlenga, Laura and Yongcheol Shin (2004). *Gravity Models of the Intra-EU Trade: Application of the Hausman–Taylor Estimation in Heterogeneous Panels with Common Time-Specific Factors*. Working Paper. University of Edinburgh, School of Economics.

Strobl, Eric and Marie-Anne Valfort (2015). “The Effect of Weather-Induced Internal Migration on Local Labor Markets: Evidence from Uganda”. In: *The World Bank Economic Review* 29.2, pp. 385–412. DOI: 10.1093/wber/lht029.

Suhrke, Astri and Sanjoy Hazarika (1993). *Pressure Points: Environmental Degradation, Migration and Conflict*. Cambridge, MA: American Academy of Arts and Sciences.

Tekin, Mehmet Kadri (2023). “Küresel İklim Değişiminin Göçler Üzerine Etkileri ve Türkiye’ye Dair Öngörüler”. Unpublished PhD Dissertation. PhD thesis. Van: Van Yüzüncü Yıl Üniversitesi, Sosyal Bilimler Enstitüsü, Coğrafya Anabilim Dalı.

Thiede, Brian, Clark Gray, and Valerie Mueller (2016). “Climate Variability and Inter-Provincial Migration in South America, 1970–2011”. In: *Global Environmental Change* 41, pp. 228–240. DOI: 10.1016/j.gloenvcha.2016.10.005.

Thorn, Jessica P. R. et al. (2023). “Exploring the Benefits and Dis-Benefits of Climate Migration as an Adaptive Strategy Along the Rural-Peri-Urban Continuum in Namibia”. In: *Regional Environmental Change* 23, p. 10. DOI: 10.1007/s10113-022-01973-5.

Tinbergen, Jan (1962). *Shaping the World Economy: Suggestions for an International Economic Policy*. New York: Twentieth Century Fund.

Tol, Richard S. J. (2002). “Welfare Specifications and Optimal Control of Climate Change: An Application of FUND”. In: *Energy Economics* 24.4, pp. 367–376. DOI: 10.1016/S0140-9883(02)00010-5.

Türkeş, Murat (2012). “Türkiye’de Gözlenen ve Öngörülen İklim Değişikliği, Kuraklık ve Çölleşme”. In: *Ankara Üniversitesi Çevrebilimleri Dergisi* 4.2, pp. 1–32. DOI: 10.1501/Csaum_0000000063.

Türkeş, Murat and Ecmel Erlat (2005). “Climatological Responses of Winter Precipitation in Turkey to Variability of the North Atlantic Oscillation During the Period 1930–2001”. In: *Theoretical and Applied Climatology* 81, pp. 45–69. DOI: 10.1007/s00704-004-0084-1.

Türkeş, Murat and Hasan Tatlı (2009). “Use of the Standardized Precipitation Index (SPI) and a Modified SPI for Shaping the Drought Probabilities over Turkey”. In: *International Journal of Climatology* 29.15, pp. 2270–2282. DOI: 10.1002/joc.1862.

Visbeck, Martin H. et al. (2001). “The North Atlantic Oscillation: Past, Present, and Future”. In: *Proceedings of the National Academy of Sciences* 98.23, pp. 12876–12877. DOI: 10.1073/pnas.231391598.

Wooldridge, Jeffrey M. (2010). *Econometric Analysis of Cross Section and Panel Data*. 2nd ed. Cambridge, MA: MIT Press.

Zouabi, Oussama (2021). “Climate Change and Climate Migration: Issues and Questions Around an In-Transition Tunisian Economy”. In: *Climatic Change* 164.3–4, p. 32. DOI: 10.1007/s10584-021-03006-2.